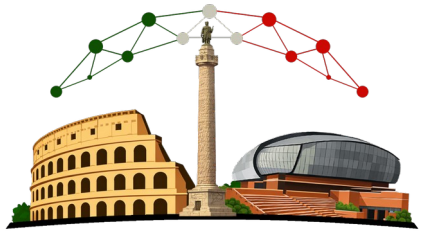
Agentic Search: Foundations, Optimization, and Long-Horizons

Chuan Meng, Litu Ou, Sean MacAvaney, and Jeff Dalton

Tutorial at the 35th ACM International Conference on Information and Knowledge Management (CIKM 2026)

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Presenters



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The University of Edinburgh



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The University of Edinburgh



Sean MacAvaney
University of Glasgow



Jeff Dalton
The University of Edinburgh

Supplementary materials

The materials, including slides, are available on the tutorial website at
https://agentic-search.github.io/



Outline

PART I 14:00–16:00

Basics and optimization

- Introduction to agentic search
- Optimization paradigms
  - No parameter updates
  - Agent learns to search
  - Search learns to serve agents
  - Agent–search joint optimization
- Live demonstration and hands-on

BREAK 16:00–16:30

PART II 16:30–18:30

Long-horizons, evaluation and beyond

- Long-horizon agentic search (deep research)
  - Reasoning-intensive QA and report generation
  - Context management
  - Efficiency
- Evaluation for agentic search
  - Evaluation for evidence and short answer
  - Evaluation for report
- Multimodal, multilingual and domain-specific
- Conclusion and open challenges

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Introduction to agentic search

I want a day trip from London on Saturday this week.
Find a seaside destination with good weather, less than two hours away by train, and a return ticket under £50.

Take 1 minute: What's your answer and strategy?

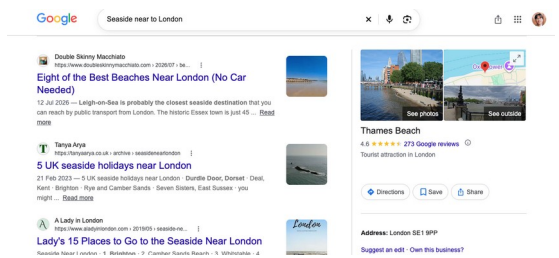
Introduction to agentic search

- **Human-driven search (traditional search): users drive much of the search process** by decomposing information needs, formulating queries, inspecting retrieved documents, and synthesizing answers

I want a day trip from London Saturday this weekend. Find a seaside destination with good weather, less than two hours away by train, and return tickets under £50. **Decompose: 1. seaside towns near London? 2. which have good weather? 3. which has a fast, cheap train?**

First check seaside towns near London

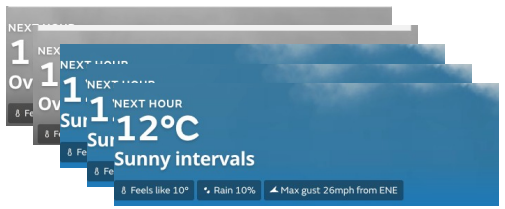
seaside towns near London



Five candidates: Brighton, Deal, Rye, Eastbourne, Durdle Door. Now check the weather

Brighton weather

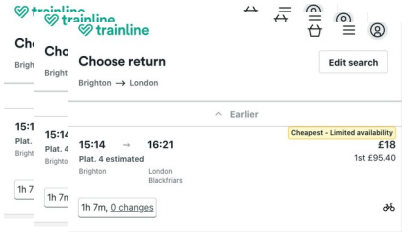
×5 towns



Only Brighton, Deal and Rye have good weather. Next, check train time and price

London to Brighton train

×3 towns



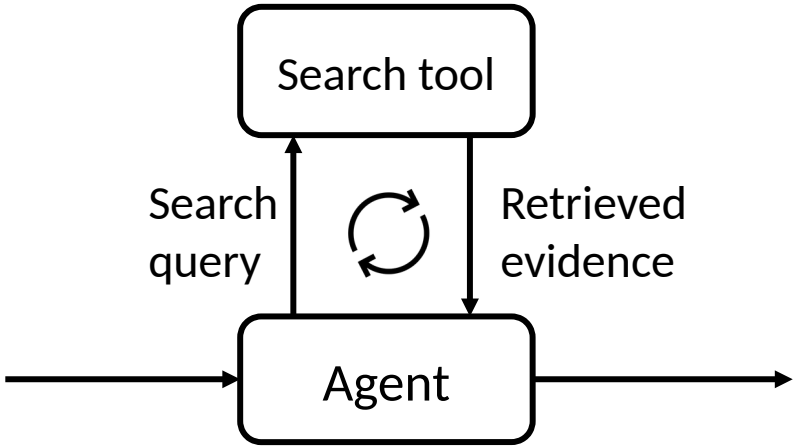
Brighton is sunny, 57 min direct, ~£36 return

Introduction to agentic search

- Agent: an autonomous system that makes decisions and takes actions on users' behalf
- **Agentic search: an agent takes over much of the cognitive burden from the user**
  - Given a complex information need: 1) generate search queries, 2) read the retrieved results, 3) loop until "enough" evidence is gathered, 4) generate an answer (with evidence)



I want a day trip from London Saturday this weekend. Find a seaside destination with good weather, less than two hours away by train, and return tickets under £50.



Evidence



Short answer

Brighton

Report

Brighton is sunny, 57 min direct, ~£36 return, with the comparison table:

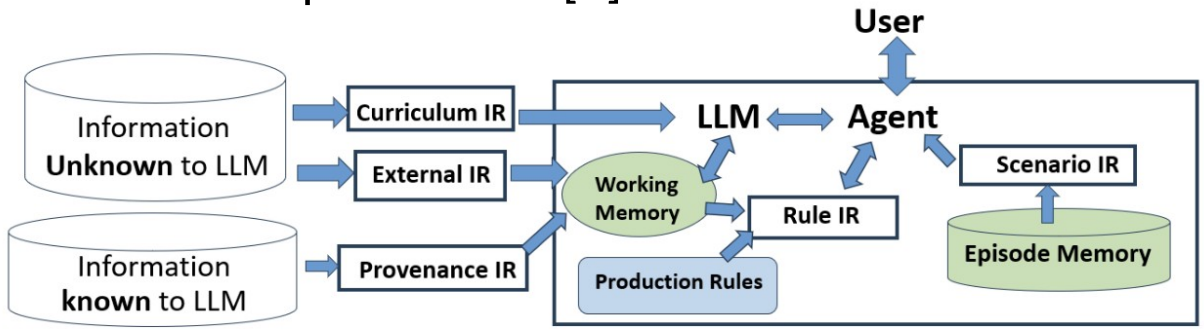
Candidates	Weather	Train	Return
Brighton	**sunny**	**57 min**	**~£36**
Deal	sunny	1 h 23	~£26
Rye	sunny	1 h 06	~£35
Eastbourne	cloudy	–	–
Durdle Door	cloudy	–	–

Introduction to agentic search

Why agentic search matters

- **For humans: it boosts productivity in knowledge-intensive work** (e.g., finance, science, policy, engineering)
  - **Hours of manual research → tens of minutes** [1]
  - It takes over searching that would otherwise cost humans valuable time, and handles needs that exceed human cognitive capacity or available time

- **For agents: search is core infrastructure**
  - **27.1%** of agent sessions on real-world user tasks call **web search** [2]
  - Agents need search to reach knowledge outside their parameters [3]
    - new or unseen information
    - the provenance of a claim
    - past cases to reuse
    - what to learn next



[1] OpenAI. Introducing Deep Research. 2025.
[2] Arena. Agent Arena: Causal Evaluation of Agents in the Real World. 2026.
[3] ChengXiang Zhai. Information Retrieval for Artificial General Intelligence: A New Perspective on Information Retrieval Research. SIGIR 2025.

Introduction to agentic search

Q & A

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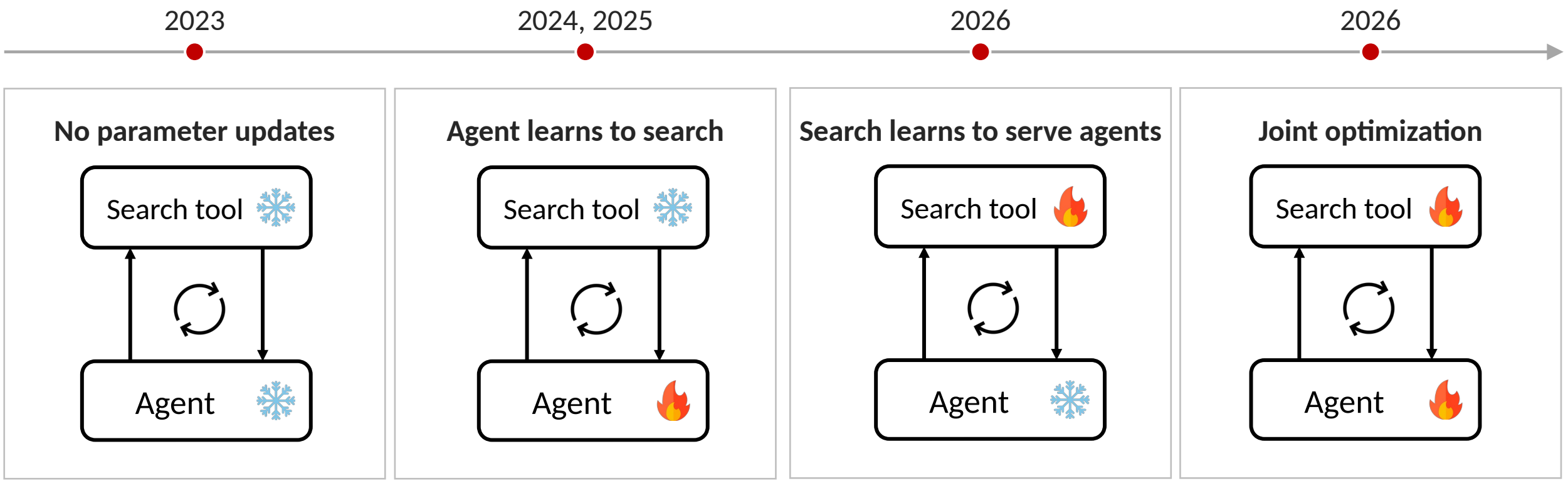
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Optimization paradigms

- Four paradigms based on which parameters are updated

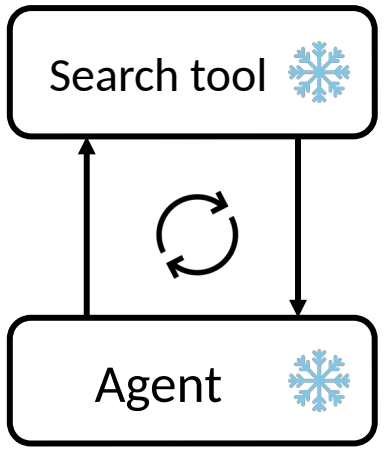


Optimization paradigms

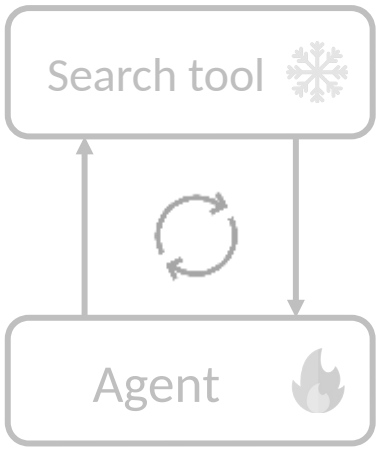
- Four paradigms based on which parameters are updated

2023 — 2024, 2025 — 2026 — 2026

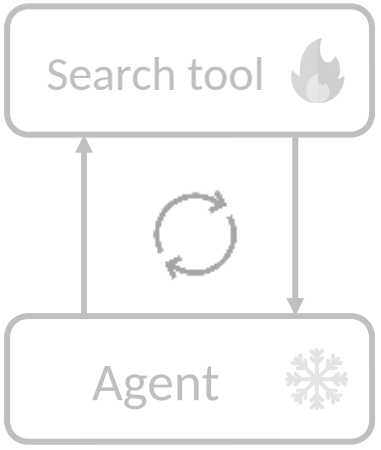
No parameter updates



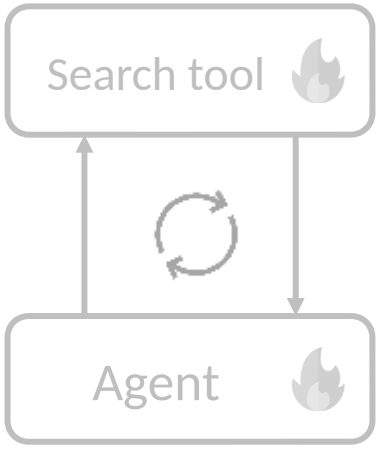
Agent learns to search



Search learns to serve agents



Joint optimization



🔥 trained ❄️ frozen

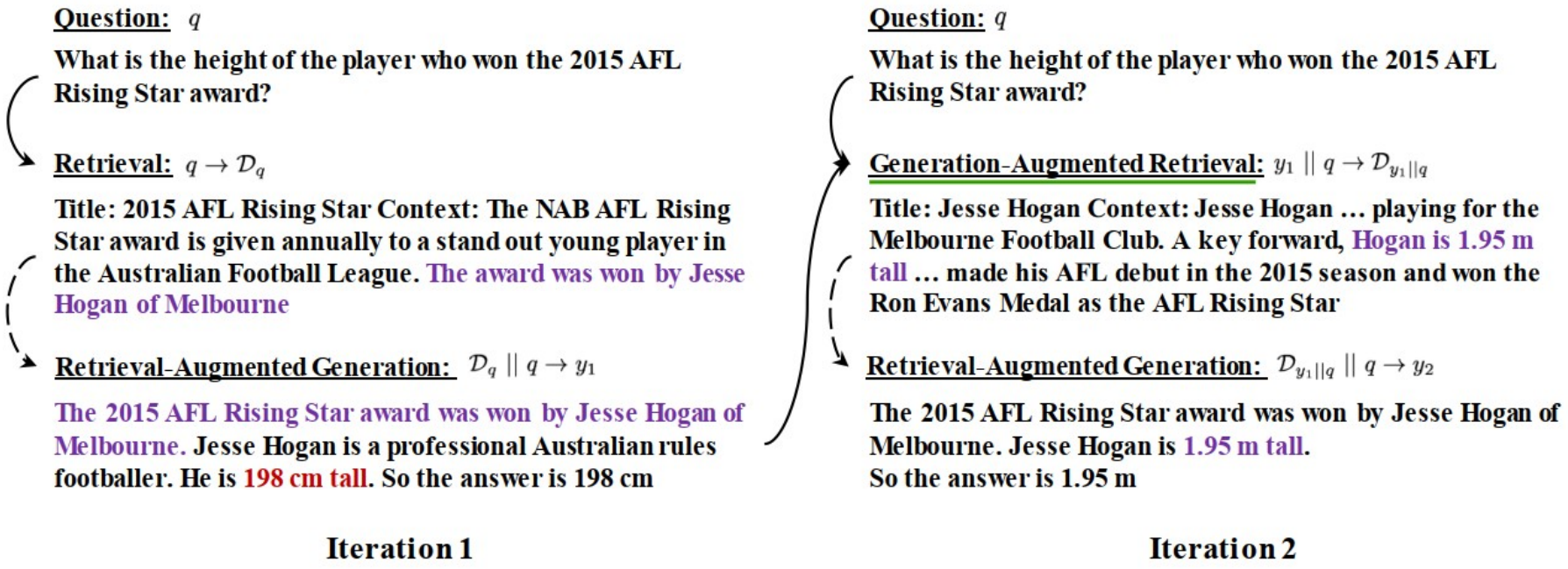
Optimization paradigms

- **No parameter updates:** prompting agents (few-shot examples) to formulate queries, invoke fixed search models, and incorporate retrieved evidence iteratively
  - Answer then search [1]
  - Analyze then search [2,3]
  - Analyze, search, then analyze again [4,5,6]

[1] Shao et al. Enhancing Retrieval-Augmented Large Language Models with Iterative Retrieval-Generation Synergy. EMNLP 2023.
[2] Jiang et al. Active Retrieval Augmented Generation. EMNLP 2023.
[3] Press et al. Measuring and Narrowing the Compositionality Gap in Language Models. EMNLP 2023.
[4] Trivedi et al. Interleaving Retrieval with Chain-of-Thought Reasoning for Knowledge-Intensive Multi-Step Questions. ACL 2023.
[5] Yao et al. ReAct: Synergizing Reasoning and Acting in Language Models. ICLR 2023.
[6] Li et al. Search-o1: Agentic Search-Enhanced Large Reasoning Models. EMNLP 2025.

Optimization paradigms

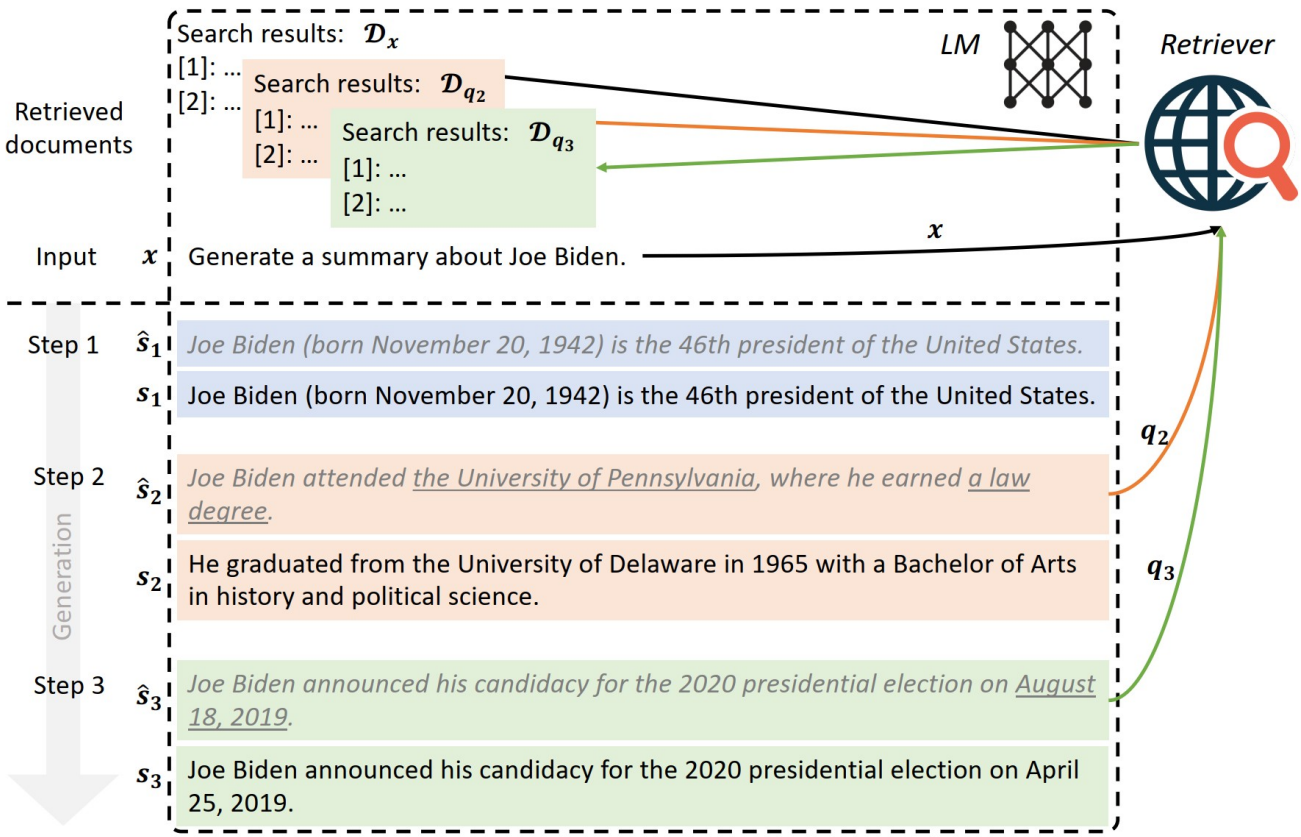
- **No parameter updates:** Answer then search
  - Iter-RetGen [1] generates an answer, uses it to guide the next search, and refines the answer
    - Search with the original question and generate an answer (might be wrong)
    - Search again with the answer and original question, and generate a new answer
    - Repeat for T iterations and return the last answer (T =2 brings most of the gains)



[1] Shao et al. Enhancing Retrieval-Augmented Large Language Models with Iterative Retrieval-Generation Synergy. EMNLP 2023.

Optimization paradigms

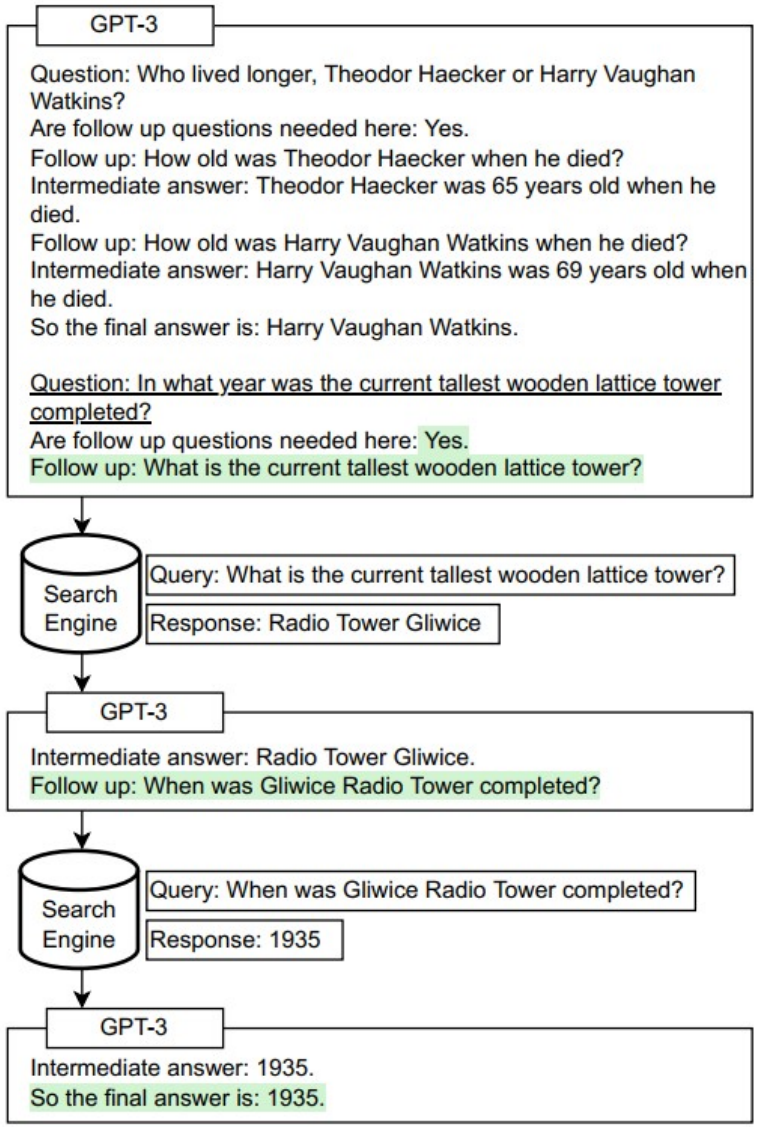
- **No parameter updates:** Analyze then search
  - FLARE [1] drafts the next sentence, checks its confidence, then searches when confidence is low
    - Generate **a draft of the next sentence**
    - If it contains low-confidence tokens, remove them and search with the remaining text
    - Use the retrieved documents to regenerate the sentence
    - Repeat until generation is complete



[1] Jiang et al. Active Retrieval Augmented Generation. EMNLP 2023.

Optimization paradigms

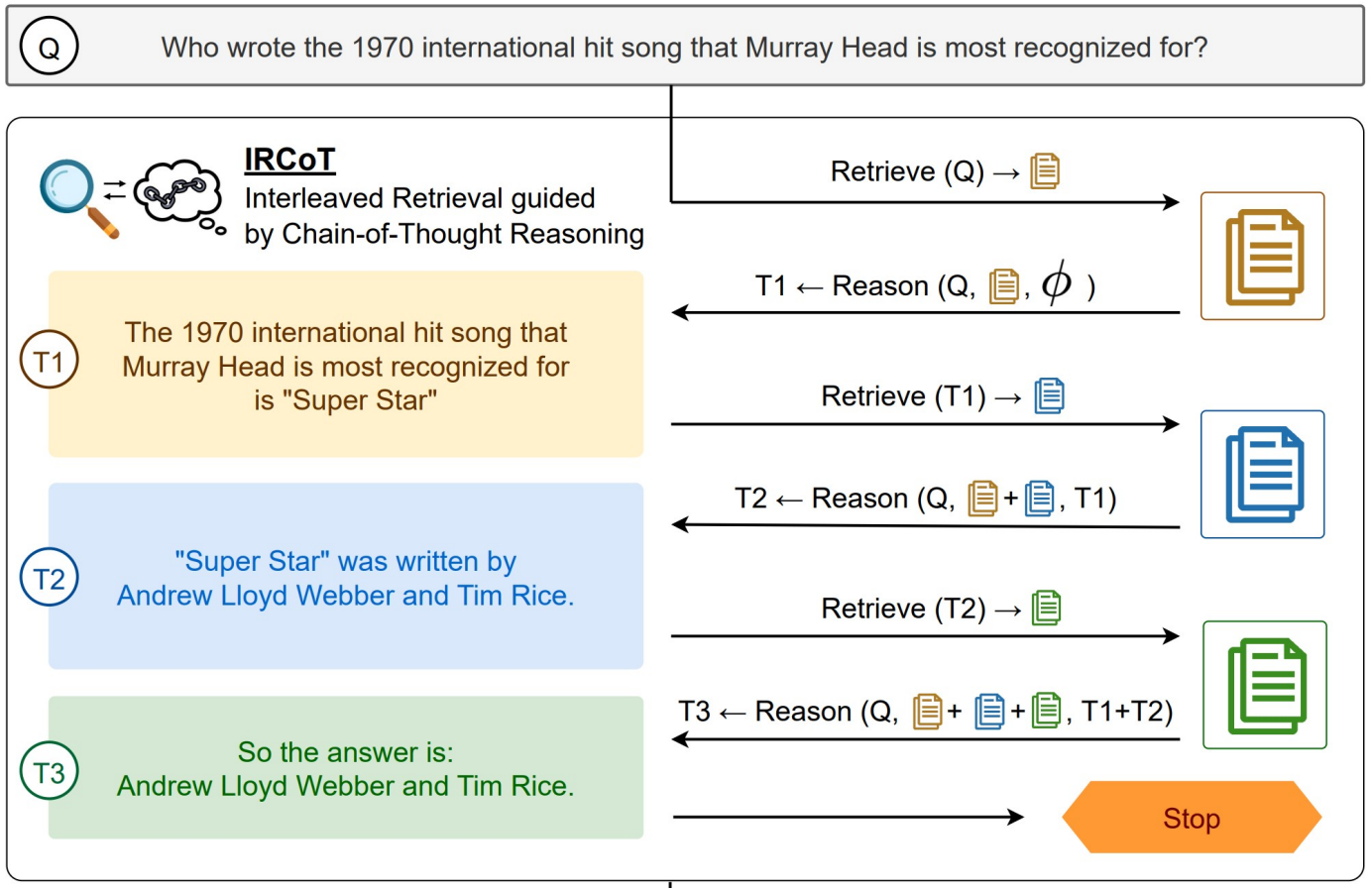
- **No parameter updates:** Analyze then search
  - Self-ask [1] identifies a needed follow-up question, then searches for its answer
    - Generate **a follow-up question** if needed
    - Search with the question
    - Insert the returned answer into the context
    - Repeat until the LLM thinks no more follow-ups are needed and gives the final answer



[1] Press et al. Measuring and Narrowing the Compositionality Gap in Language Models. EMNLP 2023.

Optimization paradigms

- **No parameter updates:** Analyze, search, then analyze again
  - IRCoT [1] first searches with the original question, generates chain-of-thought (CoT) text from the search results, and uses the generated CoT text as the next search query
    - Search with the original question
    - Use the documents to generate a CoT text
    - Search with the CoT text



[1] Trivedi et al. Interleaving Retrieval with Chain-of-Thought Reasoning for Knowledge-Intensive Multi-Step Questions. ACL 2023.

Optimization paradigms

- **No parameter updates:** Analyze, search, then analyze again
  - ReAct [1] conducts a **reasoning–acting** loop
    - **Reason to act**: Generate a reasoning trace to decide whether and what to search
    - **Act to reason**: Generate a search query, run the search, and use the results in the next reasoning step
    - Repeat until reasoning decides to stop search and generate answer, or reach T iterations

(1d) ReAct (Reason + Act)

Reasoning 1: I need to search Apple Remote and find the program it was originally designed to interact with.
Act 1 (search): Search[Apple Remote]
Retrieved evidence 1: The Apple Remote is a remote control introduced in October 2005 by Apple … originally designed to control the Front Row media center program …

Reasoning 2: Apple Remote was originally designed to control the Front Row media center program. I need to search Front Row next and find what other device can control it.
Act 2 (search): Search[Front Row]
Retrieved evidence 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports', 'Front Row (software)', …]

Reasoning 3: Front Row is not found. I need to search Front Row (software).
Act 3 (search): Search[Front Row (software)]
Retrieved evidence 3: Front Row is a discontinued media center software …

Reasoning 4: Front Row (software) is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.
Act 4 (answer): Finish[keyboard function keys] ✓

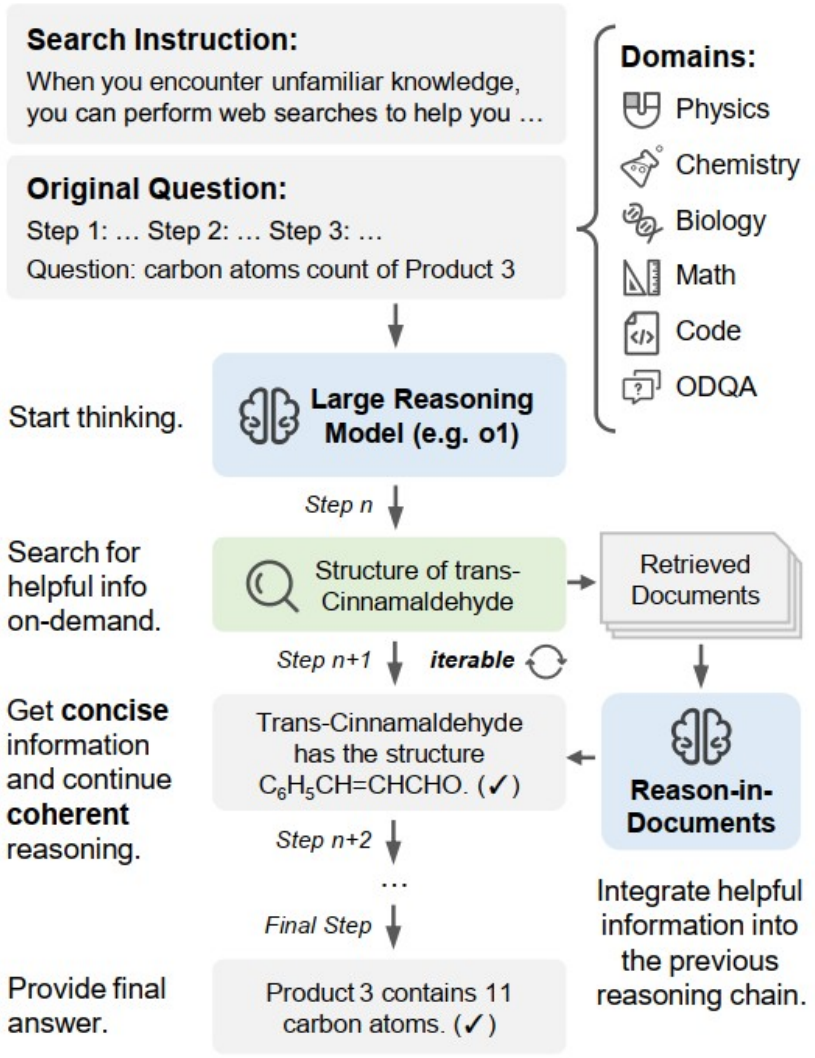
[1] Yao et al. ReAct: Synergizing Reasoning and Acting in Language Models. ICLR 2023.

Optimization paradigms

- **No parameter updates:** Analyze, search, then analyze again
  - **Previous limitation**: Verbose and noisy retrieved documents can disrupt reasoning
  - Search-o1 [1] designs a **Reason-in-Documents** module to analyze the retrieved information before injecting it into the reasoning chain

Question: When did the author of "Cogito, ergo sum" die?

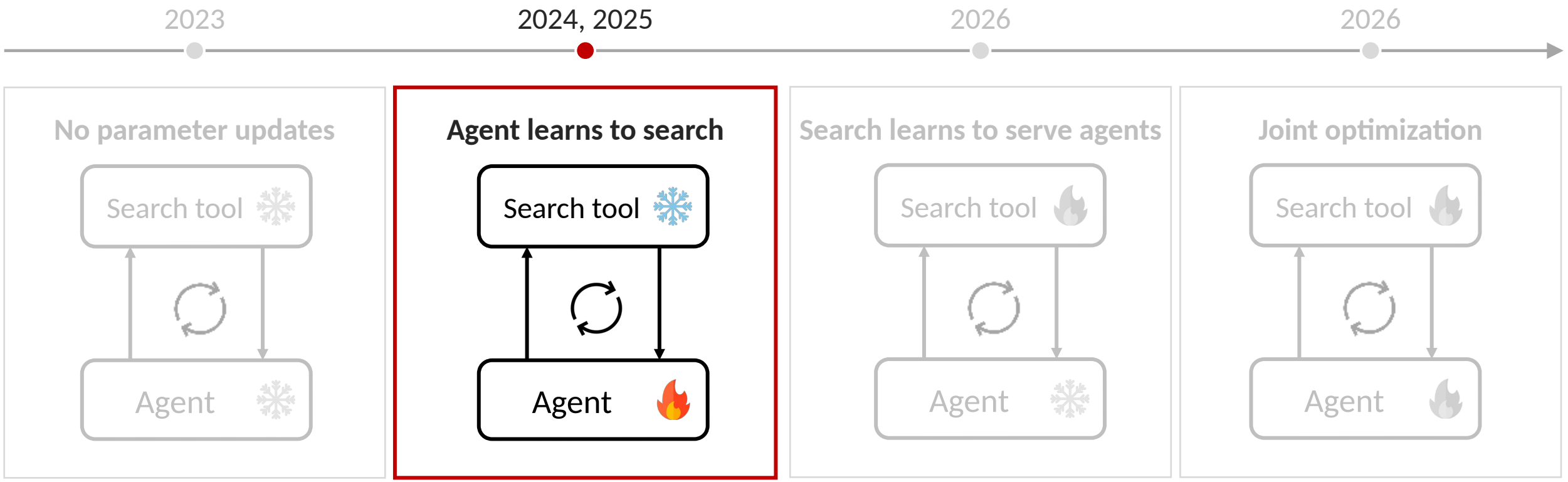
Reasoning so far: Likely Descartes. Let me confirm.
Search query: `"Cogito, ergo sum" author`
Retrieved documents
•Doc 1: The phrase appeared in a **1637** book…
•Doc 2: Descartes published *Principles of Philosophy* in **1644**…
•Doc 3: René Descartes (**1596–1650**) …
•…
Reason-in-Documents:
René Descartes, the author of "Cogito, ergo sum," lived from 1596 to 1650. His death year is **1650**



[1] Li et al. Search-o1: Agentic Search-Enhanced Large Reasoning Models. EMNLP 2025.

Optimization paradigms

- Four paradigms based on which parameters are updated



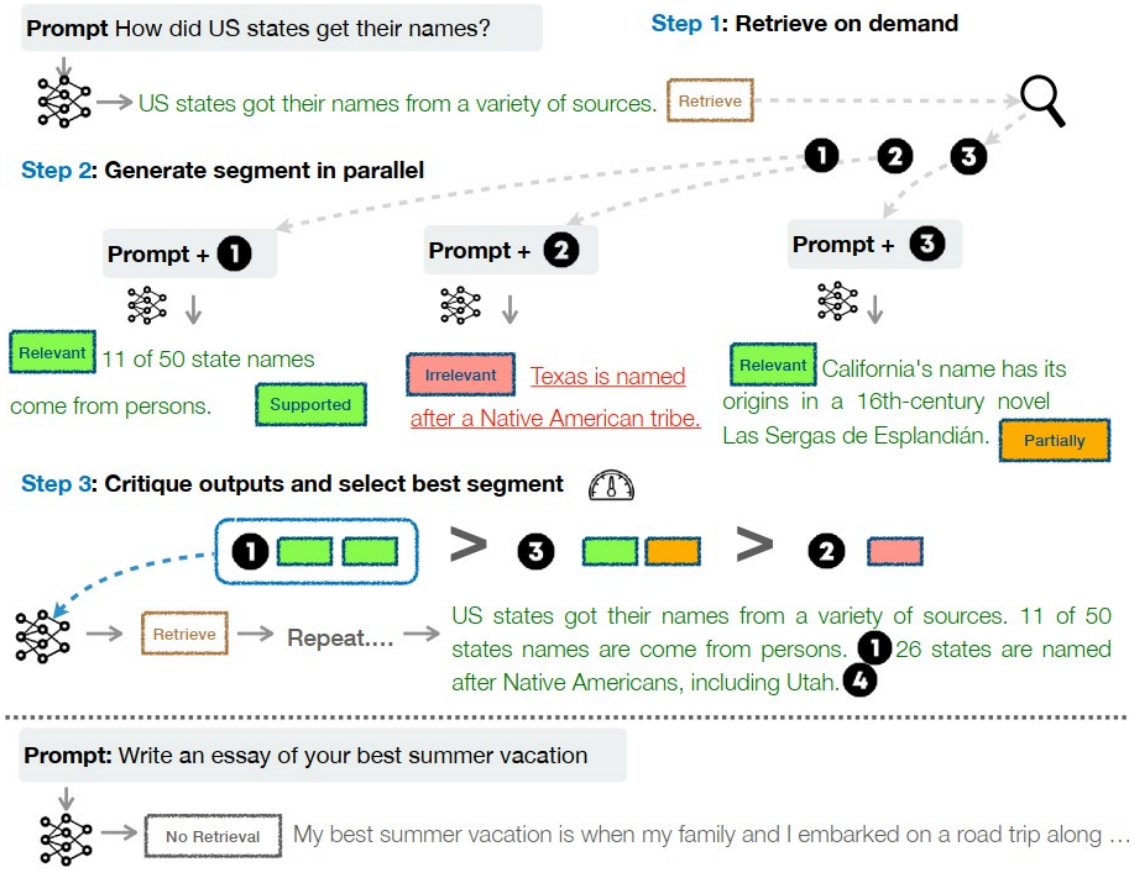
Optimization paradigms

- **Agent learns to search:** train agents to use a fixed search component
  - Learn through supervised fine-tuning (SFT) [1]
  - Learn from task rewards via reinforcement learning (RL) [2,3,4,5,6]
  - Improve RL-based agentic search
    - Training efficiency [7]
    - Inference efficiency [8]
    - Modularity and flexibility [9]

[1] Asai et al. Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. ICLR 2024.
[2] Jiang et al. DeepRetrieval: Hacking Real Search Engines and Retrievers with Large Language Models via Reinforcement Learning. COLM 2025.
[3] Jin et al. Search-R1: Training LLMs to Reason and Leverage Search Engines with Reinforcement Learning. COLM 2025.
[4] Song et al. R1-Searcher: Incentivizing the Search Capability in LLMs via Reinforcement Learning. arXiv 2025.
[5] Chen et al. ReSearch: Learning to Reason with Search for LLMs via Reinforcement Learning. NeurIPS 2025.
[6] Jin et al. An Empirical Study on Reinforcement Learning for Reasoning-Search Interleaved LLM Agents. arXiv 2025.
[7] Sun et al. ZeroSearch: Incentivize the Search Capability of LLMs without Searching. arXiv 2025.
[8] Song et al. Smart-Searcher: Incentivizing the Dynamic Knowledge Acquisition of LLMs via Reinforcement Learning. EMNLP 2025.
[9] Jiang et al. s3: You Don't Need That Much Data to Train a Search Agent via RL. EMNLP 2025.

Optimization paradigms

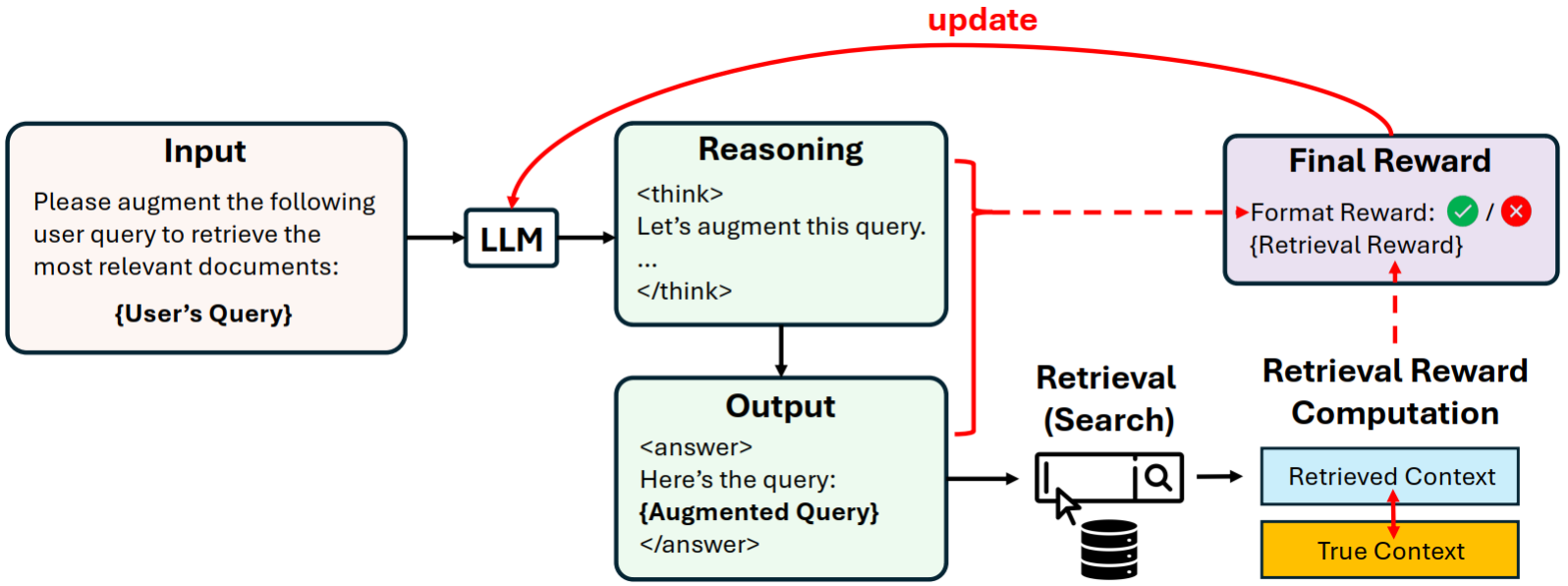
- **Agent learns to search:** Learn through SFT
  - Self-RAG [1] uses SFT to train an LLM to predict special tokens indicating 1) **retrieval necessity**, 2) **evidence relevance**, 3) **response groundedness**
    - Special tokens:
      - Retrieve/No Retrieval
      - Relevant/Irrelevant
      - Supported/Partially supported
    - SFT training steps
      - Train a critic model using GPT-4 labels
      - Use the critic to add reflection tokens to training examples
      - Fine-tune an LLM to generate reflection tokens/responses



[1] Asai et al. Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. ICLR 2024.

Optimization paradigms

- **Agent learns to search:** Learn from task rewards via RL
  - DeepRetrieval [1] trains a one-hop query augmentation model to improve retrieval effectiveness
    - Reward=retrieval reward + format reward
      - Retrieval reward: Task-specific metrics, e.g., Recall@K, NDCG, or SQL execution accuracy
      - Format reward: Check the <think>…</think><answer>…</answer> structure

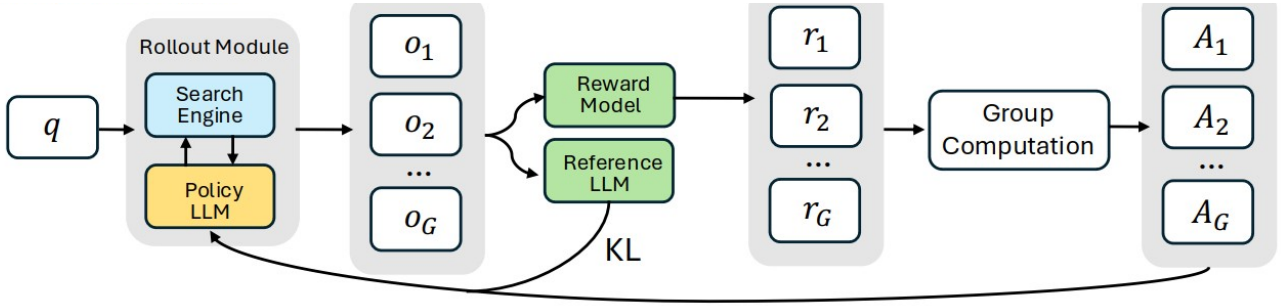


[1] Jiang et al. DeepRetrieval: Hacking Real Search Engines and Retrievers with Large Language Models via Reinforcement Learning. COLM 2025.

Optimization paradigms

- **Agent learns to search:** Learn from task rewards via RL
  - Three near-concurrent works in March 2025, namely **Search-R1** [1], **R1-Searcher** [2], and **ReSearch** [3], use RL to train LLMs to interleave reasoning and search in a ReAct-style interaction loop
    - Reward
      - Search-R1: Final-answer Exact Match (EM)
      - R1-Searcher
        - Stage 1: +0.5 for using search and +0.5 for correct format
        - Stage 2: Answer F1, with a −2 penalty for incorrect format
      - ReSearch: Answer F1; when F1 = 0, reward is 0.1 for correct format and 0 otherwise

Example: Search-R1 with Group Relative Policy Optimization (GRPO)

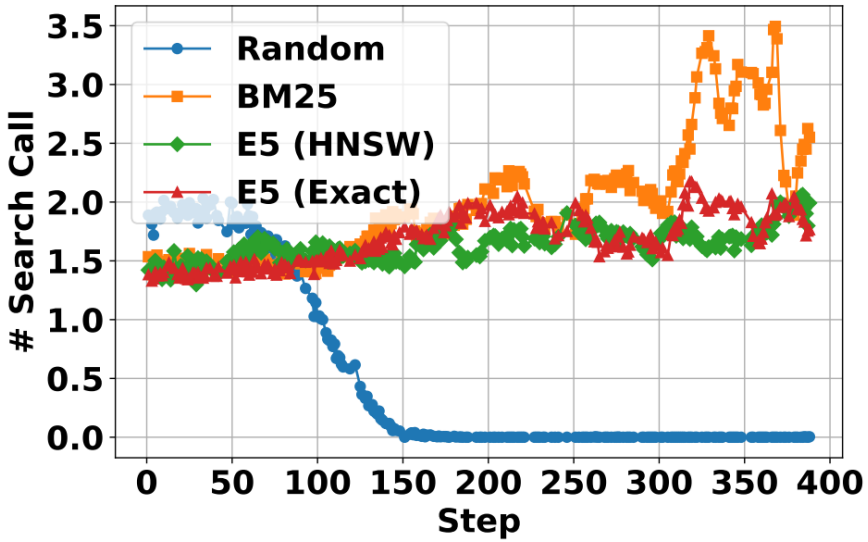


$$A_i = \frac{r_i - \text{mean}(r)}{\text{std}(r) + \epsilon}$$

[1] Jin et al. Search-R1: Training LLMs to Reason and Leverage Search Engines with Reinforcement Learning. COLM 2025.
[2] Song et al. R1-Searcher: Incentivizing the Search Capability in LLMs via Reinforcement Learning. arXiv 2025.
[3] Chen et al. ReSearch: Learning to Reason with Search for LLMs via Reinforcement Learning. NeurIPS 2025.

Optimization paradigms

- **Agent learns to search:** Learn from task rewards via RL
  - Search-R1's follow-up empirical study [1] examines reward design, model initialization, and retrieval setup
    - **Reward design**: Format rewards can help; intermediate retrieval rewards add little
    - **Initialization**: Qwen2.5-Base outperforms DeepSeek-R1-Distill-Qwen
    - **The retrieval setup shapes search behavior**
      - Random evidence → search avoidance
      - BM25 → more search calls
      - Stronger dense retrieval → more efficient search



[1] Jin et al. An Empirical Study on Reinforcement Learning for Reasoning-Search Interleaved LLM Agents. arXiv 2025.

Optimization paradigms

- **Agent learns to search**: Improve RL-based agentic search, in terms of training efficiency
  - **Previous limitation:** Frequent search calls during RL training incur high API costs; result quality is hard to control
  - **ZeroSearch** [1]: Train agents with an LLM that simulates search results
    - Simulator
      - Prompt an LLM to answer questions using Google API
      - Use an LLM judge to label retrieved results as useful or noisy
      - Fine-tune a separate LLM (simulator) to generate both types
    - Curriculum learning in RL: Gradually increase search noise

Performance in terms of EM

Search Engine	Avg.
SFT-3B	30.47
SFT-7B	33.53
SFT-14B	34.47
Google	32.81

You are the Google search engine.
Given a query, you need to generate five [useful / noisy] documents for the query.
The user is trying to answer the question: [question] whose answer is [ground truth].
Each document should contain about 30 words, and these documents should contain [useful / noisy] information.
Query: [query]
[Useful / Noisy] Output:

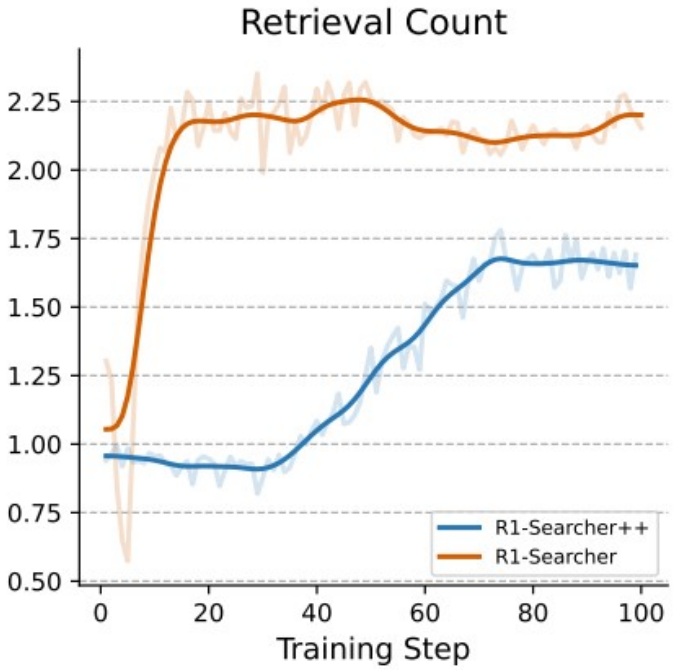
Search Engine	Queries	Training Time	GPUs Used	Total Cost
SFT-3B	~64,000	~12 hours	1 × A100 GPUs	**$17.7**
SFT-7B	~64,000	~12 hours	2 × A100 GPUs	**$35.4**
SFT-14B	~64,000	~12 hours	4 × A100 GPUs	**$70.8**
Google	~64,000	~12 hours	None	**$586.7**

[1] Sun et al. ZeroSearch: Incentivize the Search Capability of LLMs without Searching. arXiv 2025.

Optimization paradigms

- **Agent learns to search**: Improve RL-based agentic search, in terms of inference efficiency
  - **Previous limitation:** Agents overuse search, increasing inference cost
  - **R1-Searcher++** [1] encourages agents to use internal knowledge to reduce over-reliance on search
    - **Reward = answer + format + group**
      - **Answer:** 1 if the prediction covers the gold answer within 10 words, else 0
      - **Format:** 0 if valid, −2 otherwise
      - **Group:**
        - Sample multiple rollouts for the same question
        - Give a bonus to **correct rollouts with the fewest search calls**

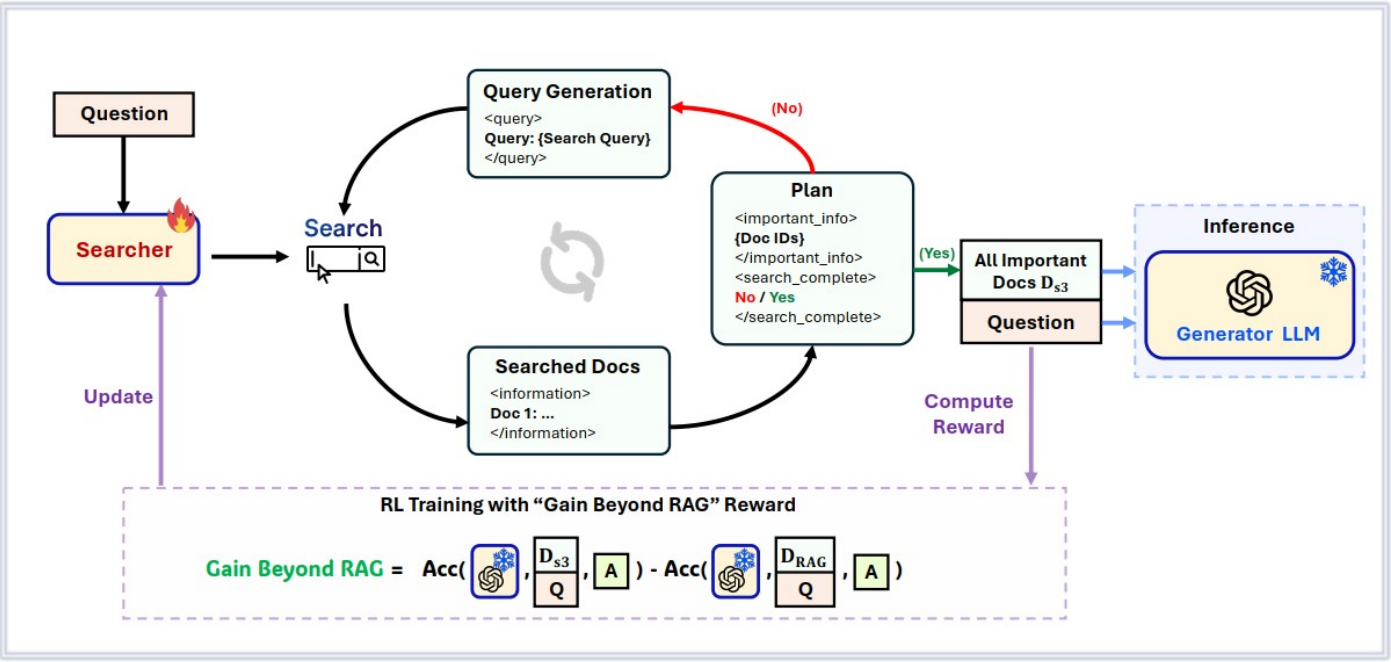
Rollout	Correct?	Search calls	Group bonus
A	Yes	**1**	**Yes**
B	Yes	3	No
C	No	0	No



[1] Song et al. Smart-Searcher: Incentivizing the Dynamic Knowledge Acquisition of LLMs via Reinforcement Learning. EMNLP 2025.

Optimization paradigms

- **Agent learns to search**: Improve RL-based agentic search, in terms of modularity and flexibility
  - **Previous limitation:** Search and answer generation are tightly coupled, making it hard to use different or proprietary generators
  - **s3** [1]: Train a separate search agent with RL, while keeping the generator frozen
    - Reward=Answer accuracy(s3 evidence) - Answer accuracy(standard RAG evidence)
    - Need fewer training steps, reducing total training time

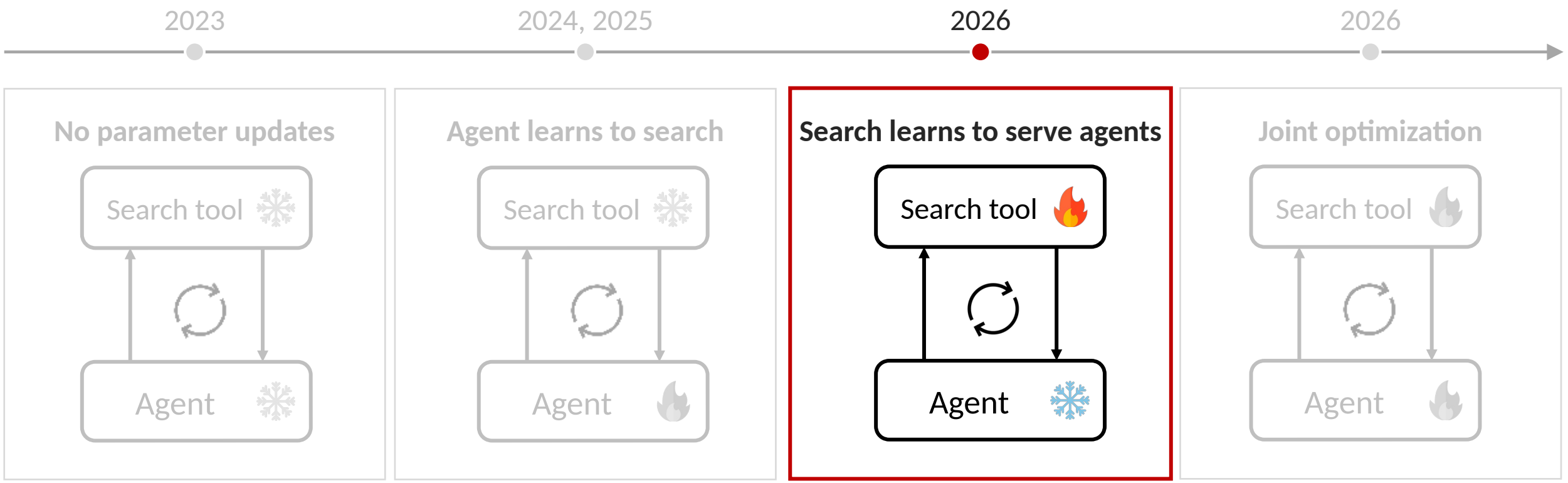


	Time/Step	Training Steps	Total
Search-R1	1.8m	~2,100	3,780m
DeepRetrieval$_{BM25}$	1.3m	~1,600	2,080m
s3	5.7m	~20	**114m**

[1] Jiang et al. s3: You Don't Need That Much Data to Train a Search Agent via RL. EMNLP 2025

Optimization paradigms

- Four paradigms based on which parameters are updated



trained frozen

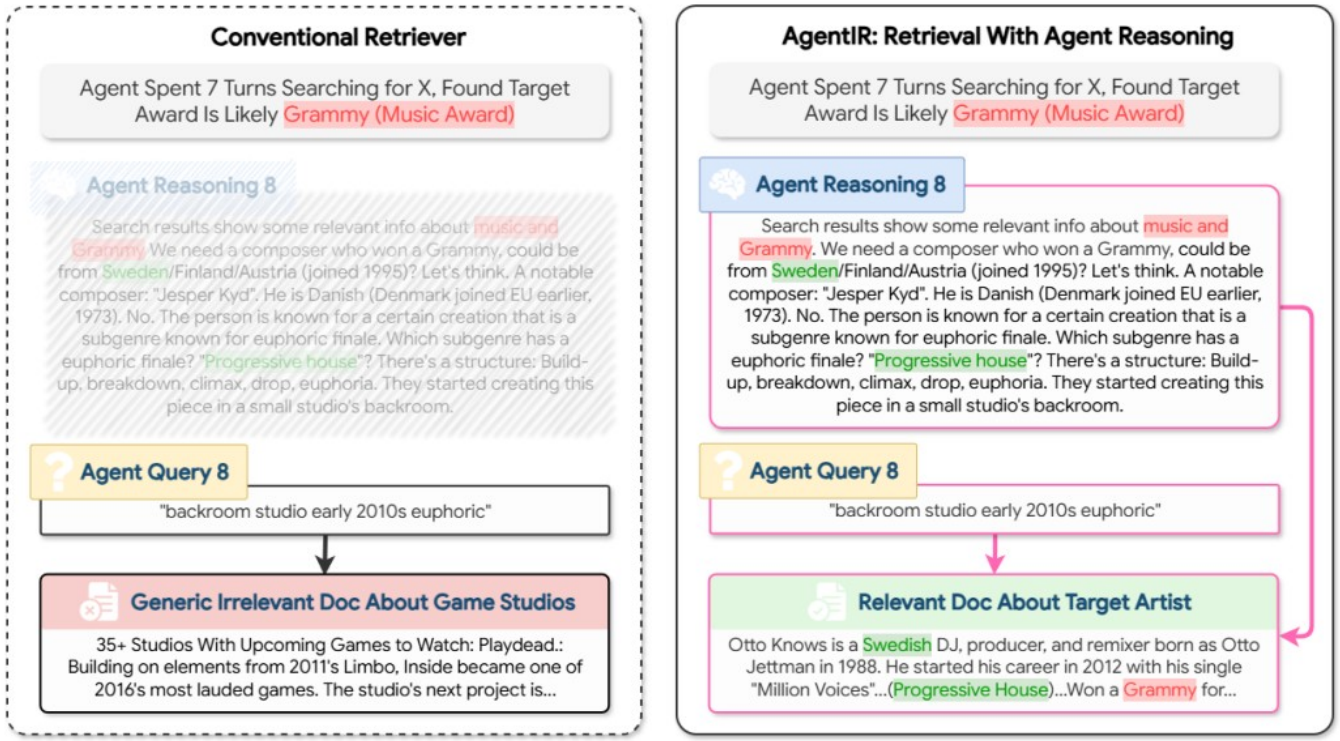
Optimization paradigms

- **Search learns to serve agents:** keeps agents fixed and trains search methods to better serve agents
  - Use agent context for search [1,2]
  - Learn from agent feedback [3]

[1] Chen et al. AgentIR: Reasoning-Aware Retrieval for Deep Research Agents. COLM 2026.
[2] Sharifymoghaddam et al. ACARe: Agent-Context-Aware Reranker for Iterative Search. arXiv 2026.
[3] Alam et al. Critic-R: Improving Agentic Search using Instruction-tuned Retrievers with Natural Language Introspective Feedback. EMNLP 2026.

Optimization paradigms

- **Search learns to serve agents**: Use agent context for search
  - **Previous limitation**: Queries alone may not fully express the agent's search intent
  - AgentIR [1] trains a dense retriever to jointly encode the query and agent's latest reasoning trace



Training procedure

- Collect reasoning–query pairs from correct agentic search trajectories on a QA dataset
- Rerank documents by an LLM given the gold answer
  - top document as positive
  - bottom documents as negatives
- Fine-tune the retriever with contrastive learning using triples: <**reasoning** + query, positive doc, negative doc>

[1] Chen et al. AgentIR: Reasoning-Aware Retrieval for Deep Research Agents. COLM 2026.

Optimization paradigms

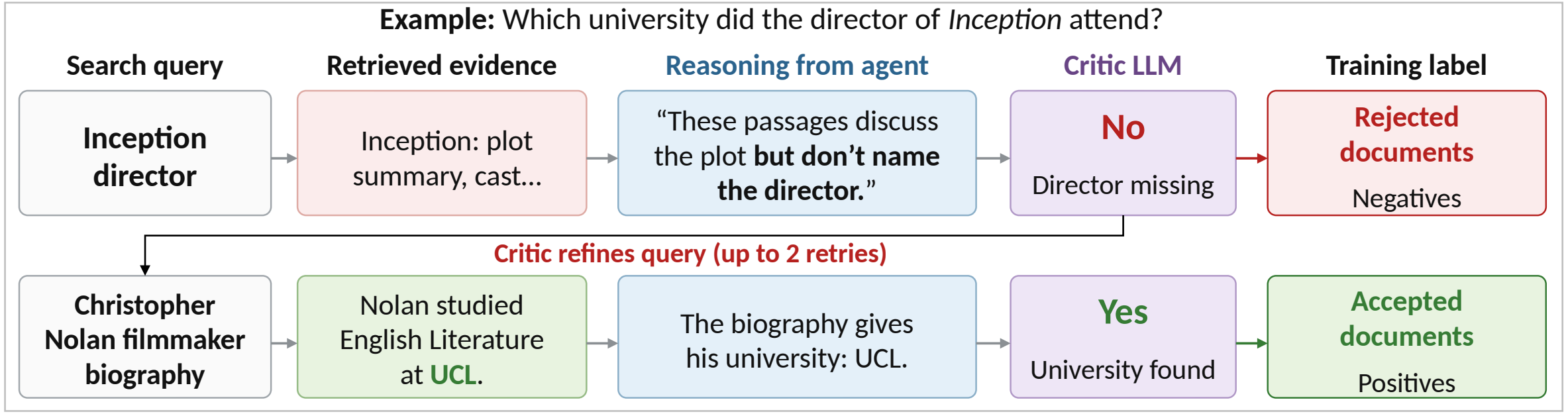
- **Search learns to serve agents**: Use agent context for search
  - ACARe [1] trains a pointwise reranker using query, original question, and agent's latest reasoning trace
    - Training procedure:
      - Reuse and expand AgentIR's synthetic training data
      - Fine-tune with contrastive learning using triples:
        - ⟨query + **original question** + **latest reasoning trace**, positive doc, negative doc⟩
      - Randomly drop the **original question** and **reasoning** independently during training

Reranker, context	nDCG@10↑	Recall@10↑	Avg. per query stats		
			Searches↓	LLM turns↓	Cost (¢)↓
No reranker	66.9 (±1.3)	68.6 (±1.0)	14.9 (±0.4)	16.0 (±0.4)	35.0 (±1.8)
Qwen3-Reranker-4B	69.7 (±1.1)†	72.2 (±1.5)†	13.1 (±0.3)†	14.1 (±0.3)†	30.8 (±0.8)†
ACARe-4B (query only)	70.8 (±0.9)†	73.4 (±1.3)†	12.7 (±0.4)†	13.8 (±0.4)†	33.0 (±2.1)
ACARe-4B (query + reasoning)	71.9 (±0.6)†*	74.9 (±0.6)†	12.2 (±0.3)†*	13.2 (±0.3)†*	31.9 (±1.3)
ACARe-4B (query + original question)	72.1 (±1.6)†	74.8 (±1.8)†	12.0 (±0.3)†*	13.0 (±0.3)†*	31.3 (±1.2)†
ACARe-4B (query + original question + reasoning)	**72.6** (±1.2)†*	**75.2** (±0.9)†*	**11.6** (±0.3)†*	**12.6** (±0.3)†*	**25.9** (±1.4)†*

[1] Sharifymoghaddam et al. ACARe: Agent-Context-Aware Reranker for Iterative Search. arXiv 2026.

Optimization paradigms

- **Search learns to serve agents**: Learn from agent feedback
  - **Previous limitation**: Human-annotated document labels are costly and may not reflect agents' reasoning needs
  - Critic-R [1] trains a retriever using document labels derived from agent feedback
    - Training procedure: A critic LLM labels retrieved evidence using the agent's reasoning

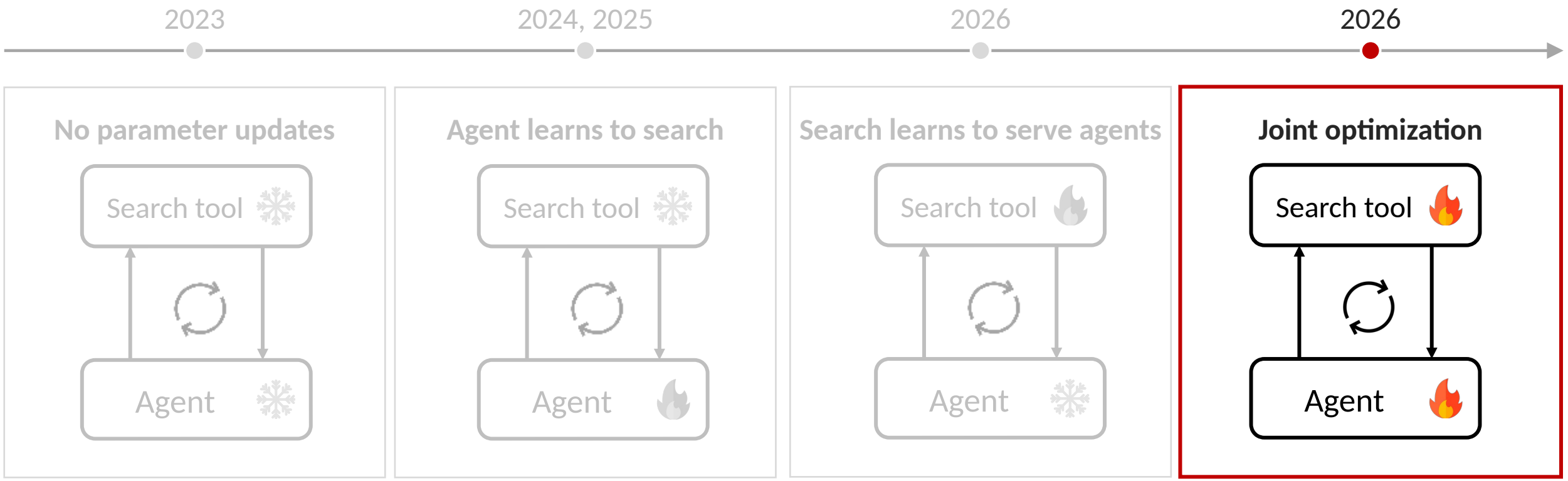


- Fine-tune retriever with contrastive learning using triples ⟨query, positive doc, negative docs⟩

[1] Alam et al. Critic-R: Improving Agentic Search using Instruction-tuned Retrievers with Natural Language Introspective Feedback. EMNLP 2026.

Optimization paradigms

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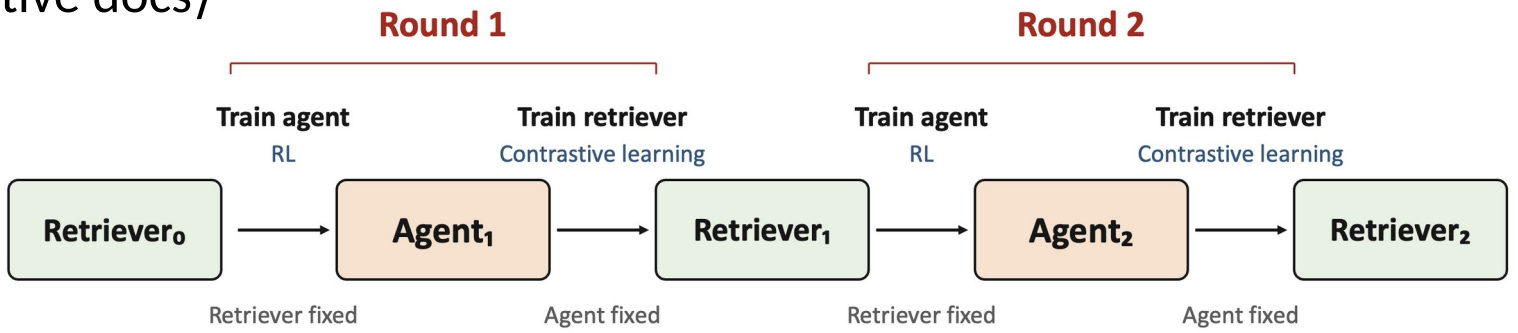
Optimization paradigms

- **Agent–search joint optimization:** jointly optimizes agents and search so that each part adapts to the other
  - Alternating training [1]
  - Simultaneous training [2]

[1] Liu et al. Agentic-R: Learning to Retrieve for Agentic Search. ACL 2026.
[2] Zeng et al. CoSearch: Joint Training of Reasoning and Document Ranking via Reinforcement Learning for Agentic Search. COLM 2026.

Optimization paradigms

- **Agent–search joint optimization**: Alternating training
  - **Previous limitation**: Optimizing only one component limits mutual adaptation
  - Agentic-R [1] alternates between optimizing the agent and the retriever
    - Training procedure (run both stages for up to two rounds)
      1. Train the agent: Follow Search-R1's RL setup, keeping the current retriever fixed
      2. Train the retriever: Keep the agent fixed
         - Generate trajectories with the improved agent
         - For each query in a trajectory, rank documents by final answer correctness, then local relevance (by an LLM); use the top doc as positive; sample lower-ranked negatives
         - Fine-tune with contrastive learning using triples: ⟨original question + query, positive doc, negative docs⟩

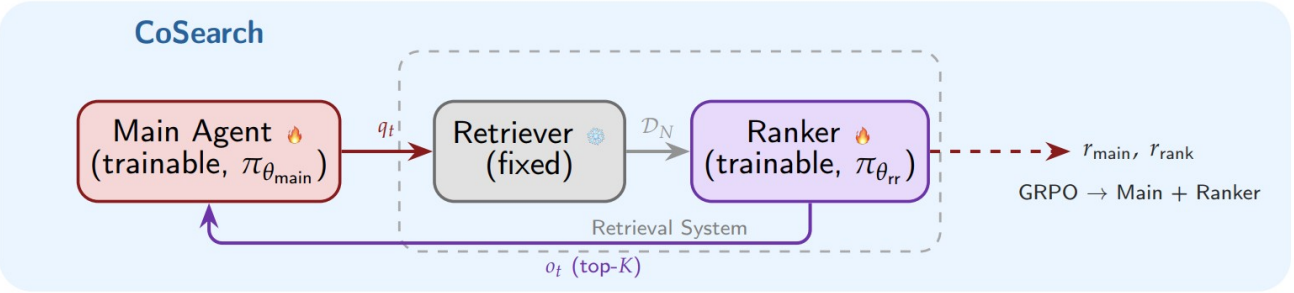


[1] Liu et al. Agentic-R: Learning to Retrieve for Agentic Search. ACL 2026.

Optimization paradigms

- **Agent–search joint optimization**: Simultaneous training
  - CoSearch [1] trains an agent and a re-ranker simultaneously via RL (GRPO)
    - Reward
      - Agent reward: final-answer F1
      - Reranker reward:

Situation	Reranker reward
A relevant doc exists but ranks low	Ranking reward (Hit@k)
A relevant doc ranks high	Ranking reward (Hit@k) + final-answer F1
No relevant doc in re-ranking candidates	Final-answer F1



Re-ranker reward ablation

Configuration	Single-Hop	Multi-Hop	Avg
CoSearch (full)	**0.619**	**0.529**	**0.568**
(a) Reward signal			
w/o r_{main}	0.612	0.508	0.560
w/o r_{rel}	0.591	0.513	0.552

[1] Zeng et al. CoSearch: Joint Training of Reasoning and Document Ranking via Reinforcement Learning for Agentic Search. COLM 2026.

Optimization paradigms

Q & A

Outline

PART I 14:00–16:00

Basics and optimization

- Introduction to agentic search
- Optimization paradigms
  - No parameter updates
  - Agent learns to search
  - Search learns to serve agents
  - Agent–search joint optimization
- **Live demonstration and hands-on**

BREAK 16:00–16:30

PART II 16:30–18:30

Long-horizons, evaluation and beyond

- Long-horizon agentic search (deep research)
  - Reasoning-intensive QA and report generation
  - Context management
  - Efficiency
- Evaluation for agentic search
  - Evaluation for evidence and short answer
  - Evaluation for report
- Multimodal, multilingual and domain-specific
- Conclusion and open challenges

Live demonstration and hands-on

- What we test
  1. **How does the agent search?**
  2. **Does the retriever matter?**
  3. **Does a re-ranker help?**
  4. **Does reading whole pages help?**
  5. **Your questions:** ask the agent anything, and we analyse what it does together

Live demonstration and hands-on

- Setup
  - **Agent:**
    - gpt-oss-20b
  - **Tools:**
    - search(query) → top 5 pages, first 512 tokens of each
    - get_document(docid) → a whole page (Step 4 only)
  - **Document corpus:** BrowseComp-Plus [1], a fixed set of about 100K web pages
  - **Question:** one question we wrote, the same in Steps 1–4; it is about December 2024, after the model's training data, so the agent has to search
    - *"Notre-Dame de Paris reopened after the 2019 fire. On what date did it reopen, and which pop star performed at the reopening?"*

[1] Chen et al. BrowseComp-Plus: A More Fair and Transparent Evaluation Benchmark of Deep-Research Agent. ACL 2026.

Live demonstration and hands-on



Ask the agent a question

- Scan the QR code, or go to **slido.com** and enter **#3678741**
- Ask anything you like, and upvote the questions you want to see. We will run some of them and go through what the agent does.
- No sign-in needed, and you can ask anonymously.

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Long-horizon agentic search: General challenges

- Characteristics of long-horizon agentic search:
  - **Longer trajectories**: Models often take tens or even hundreds of search and reasoning steps to solve a complex task.
  - **Handling Uncertainty**: Key evidence may be difficult to find, requiring agents to iteratively refine queries, explore alternative paths, backtrack from dead ends, and verify findings.
  - **Evidence management**: As information accumulates and context grows, agents must retain, organize, and synthesize relevant evidence to produce reliable answers.

- Representative Tasks:
  - **Reasoning-intensive QA:** Answer complex, well-defined questions through extensive exploration and multi-step reasoning, often requiring evidence scattered across multiple sources.
  - **Report generation:** Conduct comprehensive research on broad, open-ended topics, gathering and synthesizing information from diverse sources into structured, evidence-grounded reports.

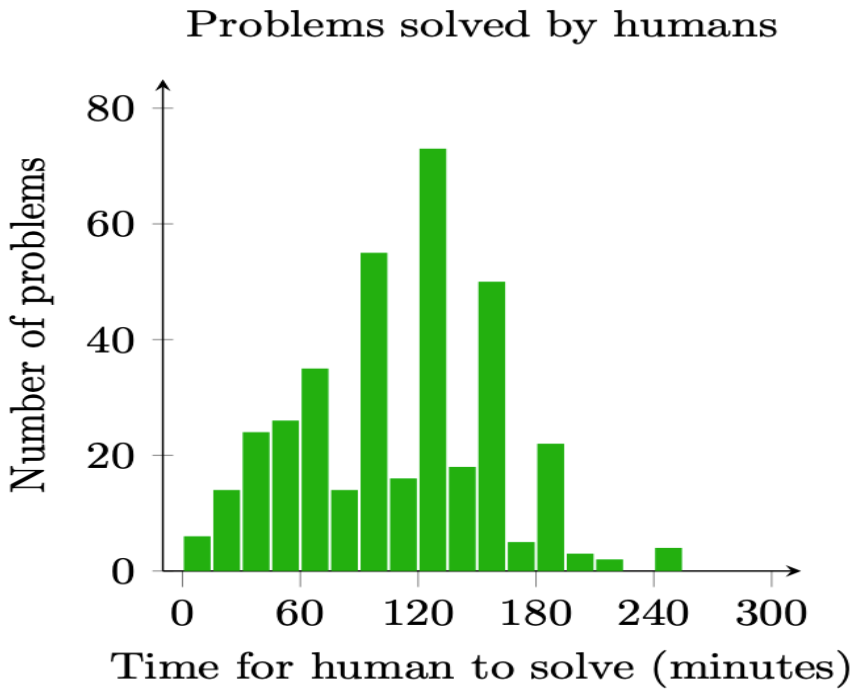
Long-horizon agentic search: Reasoning-Intensive QA

- **Reasoning-intensive QA:** four works that go from defining the task, to training agents, to evaluating them under control
  - Benchmark that defines the task: **BrowseComp [1]**
  - Training a deep research agent: **WebSailor [2]**
  - Controlled evaluation with a local corpus and reproducible retrievers: **BrowseComp-Plus [3]**
  - Large-scale ablation of retrieval and ranking design choices: **Revisiting Text Ranking [4]**

[1] Wei et al. BrowseComp: A Simple Yet Challenging Benchmark for Browsing Agents. arXiv 2025.
[2] Li et al. WebSailor: Navigating Super-human Reasoning for Web Agent. arXiv 2025.
[3] Chen et al. BrowseComp-Plus: A More Fair and Transparent Evaluation Benchmark of Deep-Research Agent. ACL 2026.
[4] Meng, Ou, MacAvaney & Dalton. Revisiting Text Ranking in Deep Research. SIGIR 2026.

Long-horizon agentic search: Reasoning-Intensive QA

- **1,266 “inverted” questions:** start from a known fact, then write a question with many constraints that point to it
- **Filtered to be hard:** unsolved by GPT-4o (with and without browsing), o1 and an early Deep Research; not on the first page of five searches; not solvable by another person within ten minutes
- **Hard to find, easy to verify:** short, single, indisputable answers, so grading is simple
- **Human trainer performance:** with no AI assistants and about two hours per question, trainers gave up on 70.8% of questions (888 / 1,255)



Wei et al. BrowseComp: A Simple Yet Challenging Benchmark for Browsing Agents. arXiv 2025.

Long-horizon agentic search: Reasoning-Intensive QA

HotpotQA: a two-hop question

Question: What government position was held by the woman who portrayed Corliss Archer in the film Kiss and Tell?

Answer: Chief of Protocol

- **Clear entities:** the question names the film and the character, so they can be searched directly
- **Fixed queries:** the reasoning path is known in advance, with few choices about what to search
- **Low information need:** a couple of supporting documents are enough to answer

BrowseComp: hard to find, easy to verify

Question (paraphrased): Which two teams played a football match between 1990 and 1994 that had a Brazilian referee, four yellow cards (two per team, three of them after half-time) and four substitutions, one of them for an injury in the first 25 minutes?

Answer: Republic of Ireland vs Romania (1990 World Cup, round of 16, referee José Roberto Wright)

- **Many weak constraints:** none names an entity to start from
- **Large candidate space:** every match 1990–94 with a Brazilian referee
- **Tens of searches:** reading match reports, rejecting and backtracking

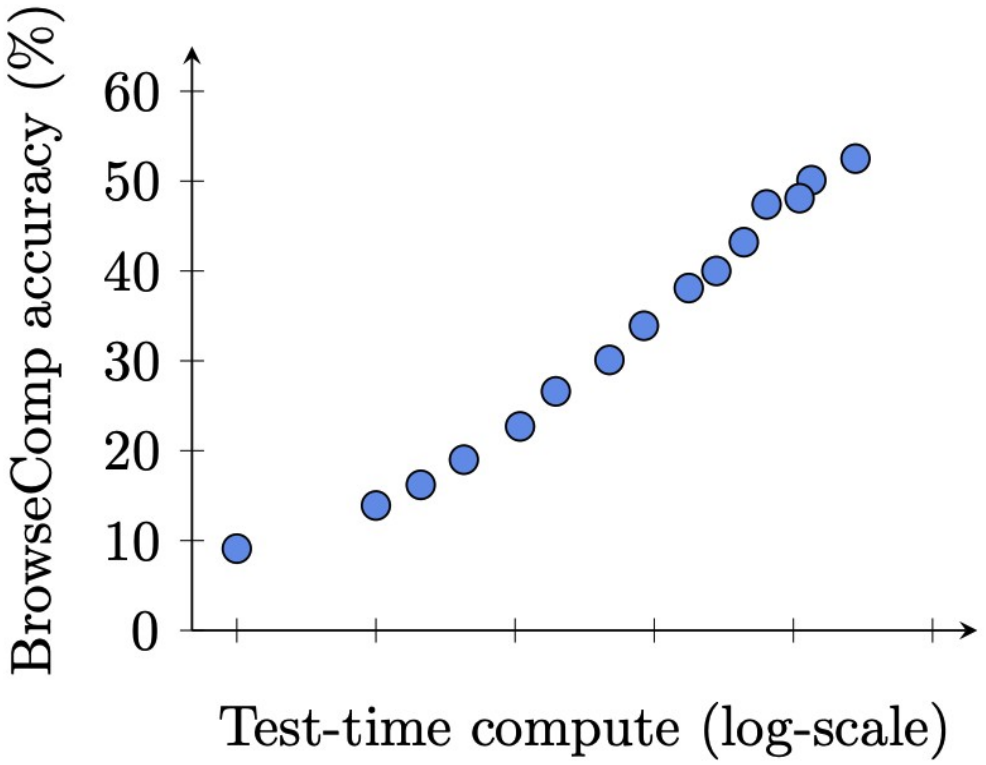
Yang et al. HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering. EMNLP 2018.
Wei et al. BrowseComp: A Simple Yet Challenging Benchmark for Browsing Agents. arXiv 2025.

Long-horizon agentic search: Reasoning-Intensive QA

Table 3: Accuracy and calibration of OpenAI models on BrowseComp.

Model	Accuracy (%)	Calibration Error (%)
GPT-4o	0.6	69
GPT-4o w/ browsing	1.9	82
GPT-4.5	0.9	68
OpenAI o1	9.9	65
Deep Research[1]	51.5	91

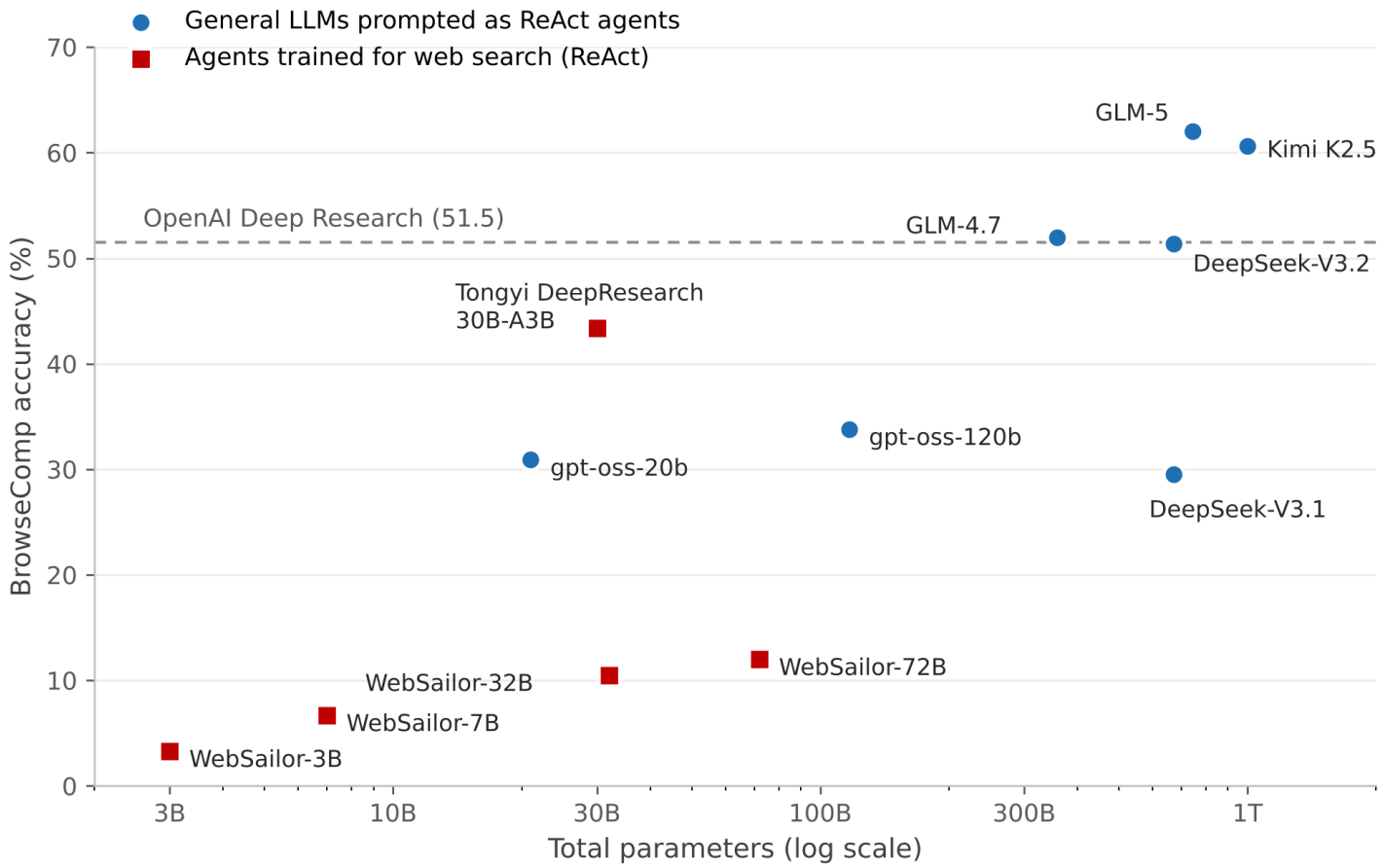
- Browsing alone is not enough: GPT-4o goes from 0.6% to only 1.9%
- Reasoning helps more than tools: o1 without browsing reaches 9.9%
- More browsing effort → higher accuracy, roughly linear in log compute
- Deep Research's large performance gap suggests the importance of dedicated training for long-horizon browsing, which earlier models largely lacked.



BrowseComp performance of an early version of OpenAI Deep Research with varying browsing effort. Accuracy scales smoothly with test-time compute.

Wei et al. BrowseComp: A Simple Yet Challenging Benchmark for Browsing Agents. arXiv 2025.

Long-horizon agentic search: Reasoning-Intensive QA



Takeaways

- Several large open-source models now pass the original Deep Research performance
- Tongyi DeepResearch (30B, 43.4%) outperforms the much larger DeepSeek-V3.1 (671B, 29.5%), highlighting the potential importance of specialized training.
- Strong performance with smaller models remains difficult, leaving considerable room for more effective and efficient search agents.

* Results gathered from official model reports and published papers (single ReAct run on the live web; evaluation setups differ across sources).

Long-horizon agentic search: Reasoning-Intensive QA

- **Reasoning-intensive QA:** four works that go from defining the task, to training agents, to evaluating them under control
  - Benchmark that defines the task: BrowseComp [1]
  - **Training a deep research agent: WebSailor [2]**
  - Controlled evaluation with a local corpus and reproducible retrievers: BrowseComp-Plus [3]
  - Large-scale ablation of retrieval and ranking design choices: Revisiting Text Ranking [4]

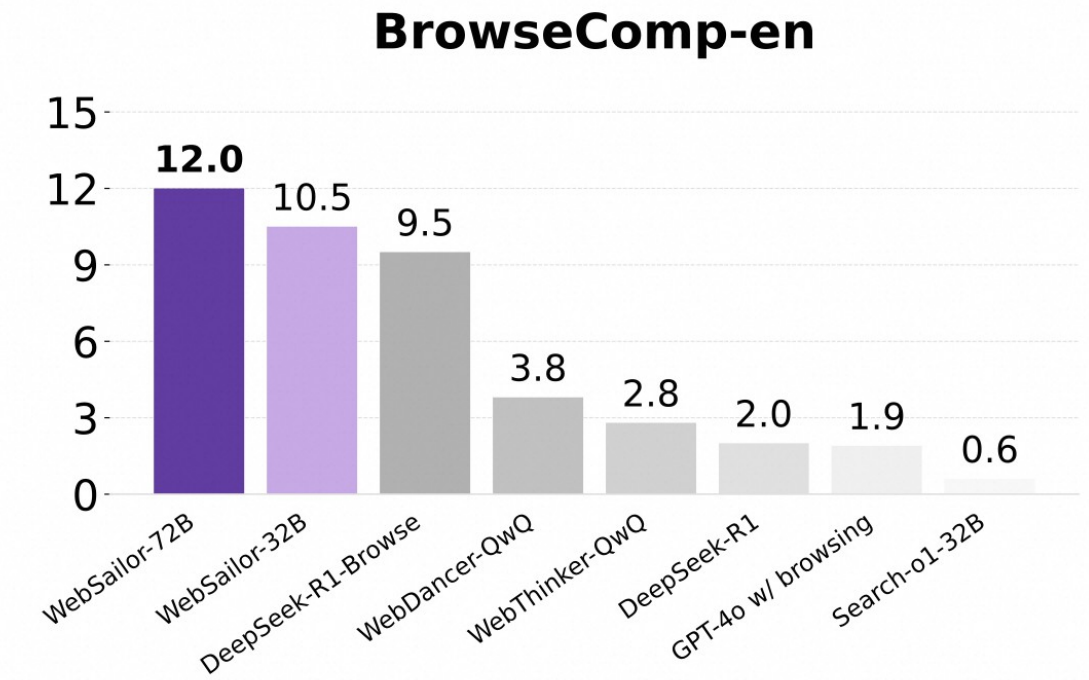
BrowseComp *defines the task: questions that are hard to find but easy to verify. Its results also suggest that dedicated training for long-horizon browsing matters.*

Next: how to train such an agent, with WebSailor

[1] Wei et al. BrowseComp: A Simple Yet Challenging Benchmark for Browsing Agents. arXiv 2025.
[2] Li et al. WebSailor: Navigating Super-human Reasoning for Web Agent. arXiv 2025.
[3] Chen et al. BrowseComp-Plus: A More Fair and Transparent Evaluation Benchmark of Deep-Research Agent. ACL 2026.
[4] Meng, Ou, MacAvaney & Dalton. Revisiting Text Ranking in Deep Research. SIGIR 2026.

Long-horizon agentic search: Reasoning-Intensive QA

- **Hypothesis:** strong deep research agents share one skill, systematically reducing extreme uncertainty across a vast information landscape; WebSailor is an end-to-end recipe to train it into open models
- **Synthetic data**
  - SailorFog-QA: Level 3 questions from random walks over rare entities, then obfuscated
  - Reasoning reconstruction: expert action–observation traces with short, rewritten thoughts
- **Training**
  - RFT cold start: ~2k filtered trajectories teach tool use and long-horizon structure
  - DUPO: agentic RL with duplicating sampling, 2–3× faster than DAPO

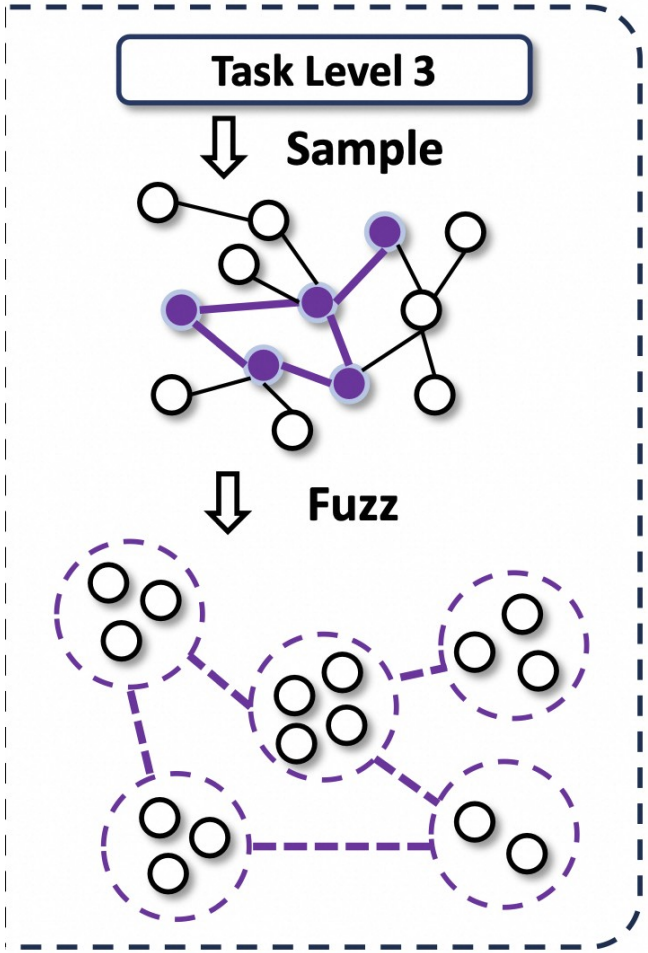


State-of-the-art performance against open-source models and agents at the time

Li, Zhang, Yin, Zhang, Ou et al. WebSailor: Navigating Super-human Reasoning for Web Agent. arXiv 2025.

Long-horizon agentic search: Reasoning-Intensive QA

- Three levels of information-seeking tasks
  - Level 1: low uncertainty; internal knowledge or one search
  - Level 2: high initial uncertainty but a clear path (multi-hop QA)
  - Level 3: high uncertainty that is hard to reduce; entities coupled in complex ways with no predefined reasoning path
- Shortcoming: existing training data stops at Level 2 (on WebDancer-QA, over half of expert trajectories need only 2 tool calls), while BrowseComp is dominated by Level 3
- SailorFog-QA: synthesised Level 3 questions
  - Sample: random walks from rare Wikidata entities grow dense entity graphs; sample subgraphs of coupled entities and write a question for each
  - Fuzz: obfuscate the clues (“in the early 2010s”, “a market share below 1%”) so no single lookup resolves them



Li, Zhang, Yin, Zhang, Ou et al. WebSailor: Navigating Super-human Reasoning for Web Agent. arXiv 2025.

Long-horizon agentic search: Reasoning-Intensive QA

- **Supervision data:** QwQ-32B solves SailorFog-QA; only successful traces are kept, and raw LRM thoughts are too verbose to imitate
- **Cold-start data:** just over 2k trajectories (correct answer, under 32k tokens, more than 5 tool calls); observations masked in the loss
- **RL reward:** 0.1 × format + 0.9 × answer (LLM judge), with GRPO-style group advantages and token-level loss

Reasoning reconstruction

Keep the expert's actions and observations, but rewrite each step's thought to be short and concise

→

Stage 1: RFT cold start

Fine-tune on a small set of filtered, successful trajectories, so the agent learns tool use before RL

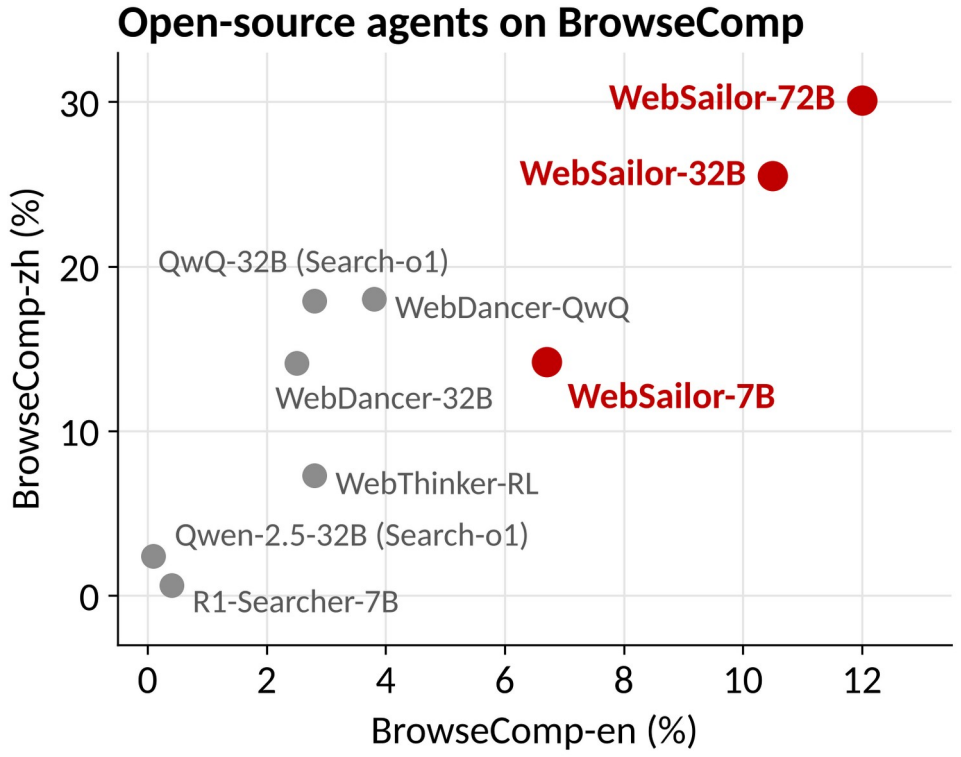
→

Stage 2: DUPO RL

Drop questions that are too easy; fill each batch by duplicating groups with mixed outcomes, 2–3× faster than DAPO

Li, Zhang, Yin, Zhang, Ou et al. WebSailor: Navigating Super-human Reasoning for Web Agent. arXiv 2025.

Long-horizon agentic search: Reasoning-Intensive QA



Model	SimpleQA pass@1 (%)
Agents with search	
WebSailor-72B	**93.5**
WebSailor-32B	**92.8**
WebDancer-QwQ	90.5
WebThinker-RL	77.5
DeepSeek-R1 (ReAct)	72.2
Direct inference	
GPT-4.1	41.6
GPT-4o	38.2
DeepSeek-R1	27.8

WebSailor is the best open-source agent on BrowseComp

Performance generalises to simpler tasks: single-fact lookup on SimpleQA

Li, Zhang, Yin, Zhang, Ou et al. WebSailor: Navigating Super-human Reasoning for Web Agent. arXiv 2025.
Wei et al. Measuring Short-form Factuality in Large Language Models (SimpleQA). arXiv 2024.

Long-horizon agentic search: Reasoning-Intensive QA

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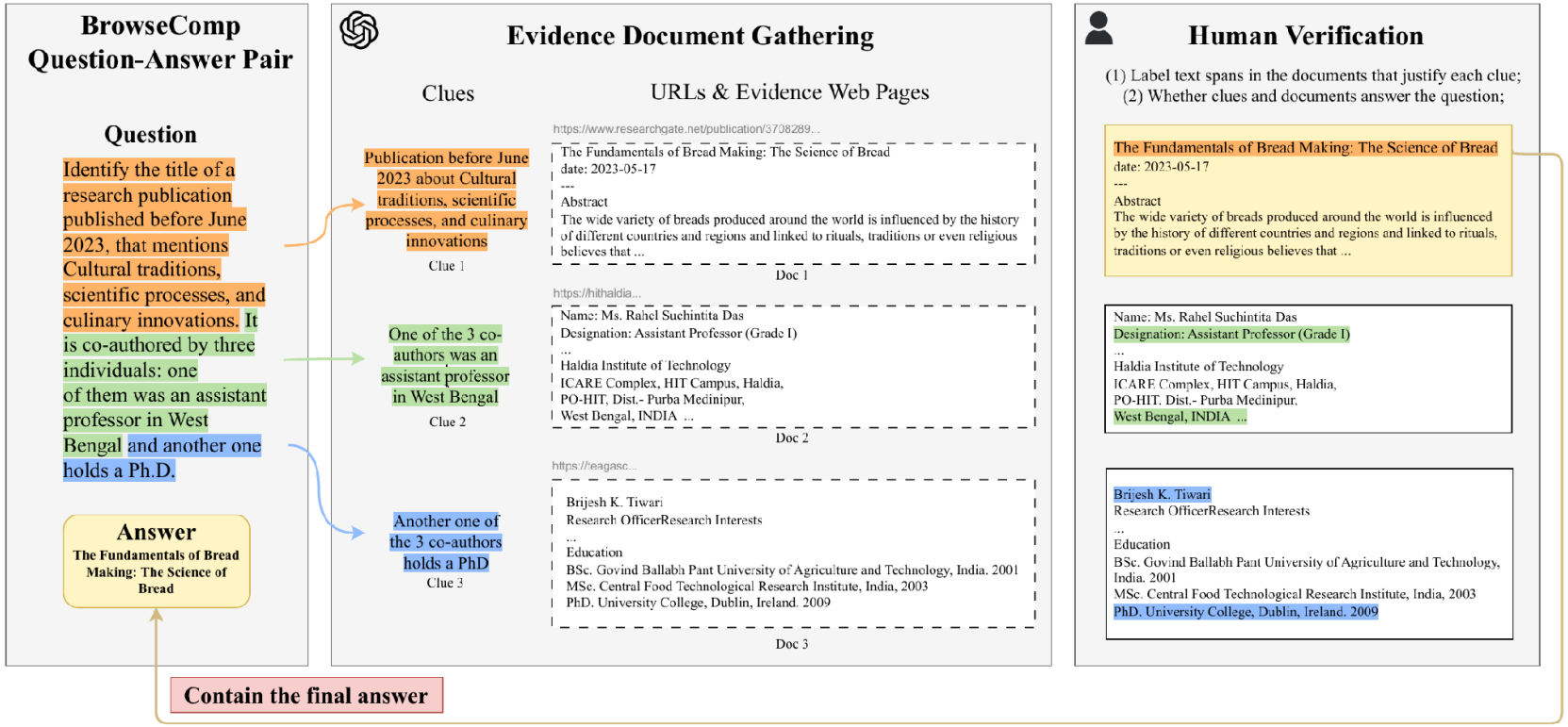
*So far, both **BrowseComp** and **WebSailor** are evaluated on the live web, where search results are opaque and change over time.*

Next: a controlled, reproducible setup with BrowseComp-Plus

[1] Wei et al. BrowseComp: A Simple Yet Challenging Benchmark for Browsing Agents. arXiv 2025.
[2] Li et al. WebSailor: Navigating Super-human Reasoning for Web Agent. arXiv 2025.
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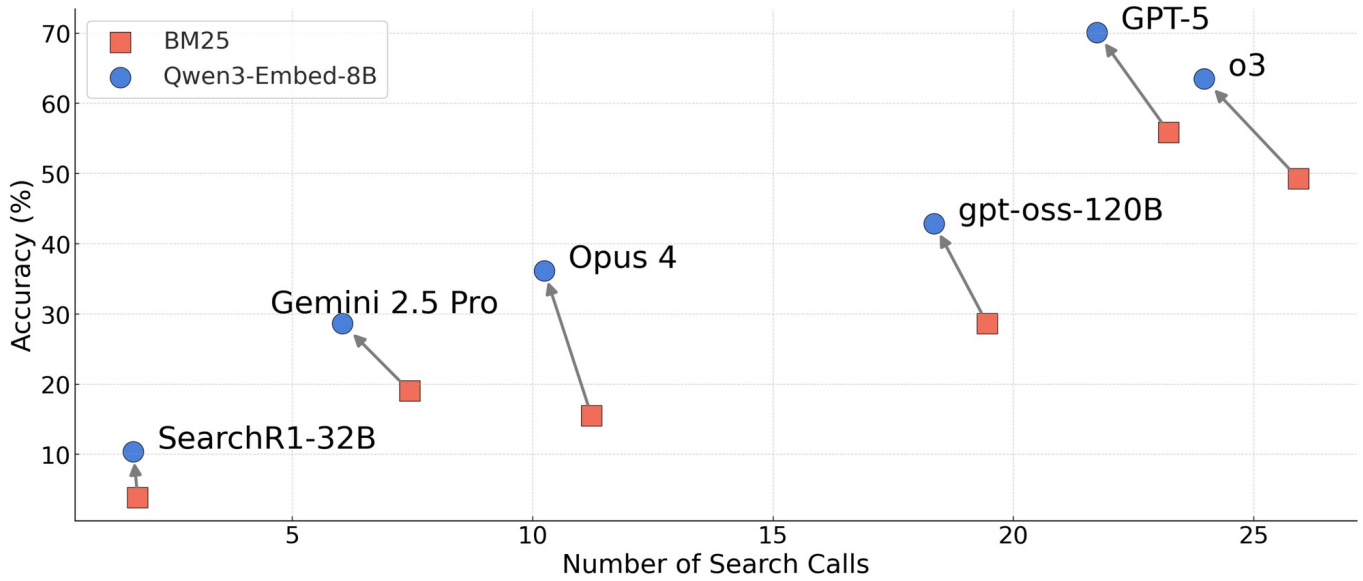
- 830 BrowseComp questions, each human-verified with answer-supporting evidence
- Fixed corpus of 100,195 documents: evidence, gold documents and hard negatives
- Human relevance labels let us evaluate the retriever separately from the agent



Evidence collection and human verification

Long-horizon agentic search: Reasoning-Intensive QA

- Same BM25 retriever, different agents
  - Search-R1: 3.9% vs GPT-5: 55.9%
- Same GPT-5 agent, different retrievers
  - BM25: 55.9% accuracy, 23.2 calls
  - Qwen3-Embed-8B: 70.1% accuracy, 21.7 calls
- Both the agent and the retriever matter



Accuracy vs. number of search calls, BM25 (red) vs Qwen3-Embed-8B (blue)

Chen et al. BrowseComp-Plus: A More Fair and Transparent Evaluation Benchmark of Deep-Research Agent. ACL 2026.

Long-horizon agentic search: Reasoning-Intensive QA

- **Reasoning-intensive QA:** four works that go from defining the task, to training agents, to evaluating them under control
  - Benchmark that defines the task: BrowseComp [1]
  - Training a deep research agent: WebSailor [2]
  - Controlled evaluation with a local corpus and reproducible retrievers: BrowseComp-Plus [3]
  - **Large-scale ablation of retrieval and ranking design choices: Revisiting Text Ranking [4]**

BrowseComp-Plus *shows that the retriever alone matters: with GPT-5 fixed, a stronger retriever lifts accuracy from 55.9% to 70.1%.*

Next: a larger-scale ablation of which retrieval and ranking choices matter

[1] Wei et al. BrowseComp: A Simple Yet Challenging Benchmark for Browsing Agents. arXiv 2025.
[2] Li et al. WebSailor: Navigating Super-human Reasoning for Web Agent. arXiv 2025.
[3] Chen et al. BrowseComp-Plus: A More Fair and Transparent Evaluation Benchmark of Deep-Research Agent. ACL 2026.
[4] Meng et al. Revisiting Text Ranking in Deep Research. SIGIR 2026.

Long-horizon agentic search: Reasoning-Intensive QA

Revisiting Text Ranking in Deep Research

- Deep research agents usually search through black-box web search APIs
- So it is unclear how established text ranking methods behave when the searcher is an LLM agent
- Replace the API with transparent retrievers and re-rankers; tested on **BrowseComp-Plus**

Three research questions

- **RQ1 Retrievers:** which retrievers work best for agent-issued queries, on passages and documents?
- **RQ2 Re-ranking:** does re-ranking still help when the reader is an agent?
- **RQ3 Query mismatch:** agents issue keyword queries, rankers are trained on questions. Does this hurt?

Meng et al. Revisiting Text Ranking in Deep Research. SIGIR 2026.
Chen et al. BrowseComp-Plus: A More Fair and Transparent Evaluation Benchmark of Deep-Research Agent. ACL 2026.

Long-horizon agentic search: Reasoning-Intensive QA

RQ1 BM25 performs best on passages

Contrary to BrowseComp-Plus, where BM25 on full documents was the weakest retriever

Answer accuracy (%), gpt-oss-20b agent

Retriever	Passages	Documents
BM25	**57.2**	25.9
ColBERTv2	52.1	48.1
SPLADE-v3	51.6	47.6
Qwen3-Embed-8B	41.7	42.1
RepLLaMA	40.6	36.3

Agent queries look like web-search keywords

Agent	Search query
gpt-oss-20b	“90+7” attendance 61700 “Man United” “4-1” “90+4”
GLM-4.7-Flash	“61,888” football match Stockholm Vienna Prague “goal in the 6th minute” football

- Keyword and quoted-phrase queries suit lexical matching (and multi-vector ColBERTv2)
- BM25 is weak on documents because of length normalisation: indexing only the first 512 tokens doubles accuracy (**25.9 → 51.3**)

Meng et al. Revisiting Text Ranking in Deep Research. SIGIR 2026.
Chen et al. BrowseComp-Plus: A More Fair and Transparent Evaluation Benchmark of Deep-Research Agent. ACL 2026.

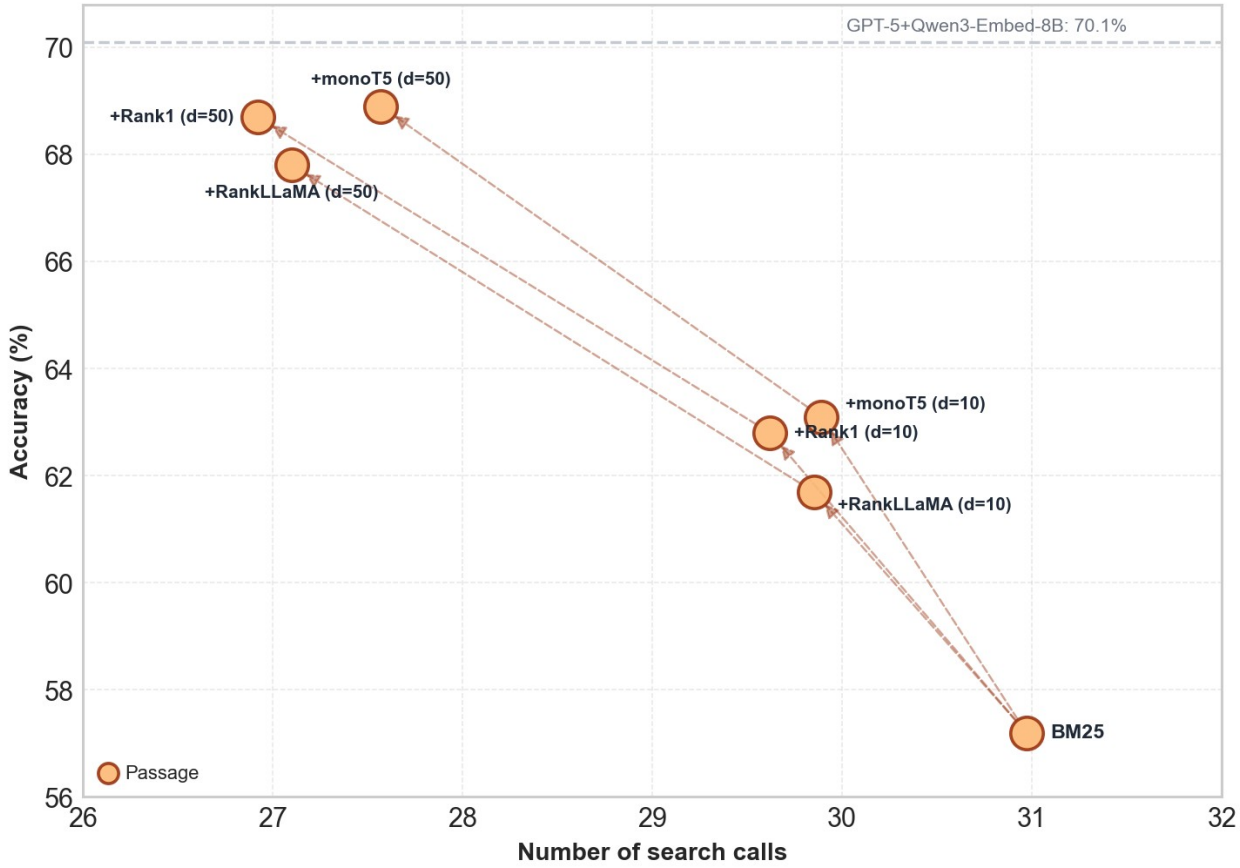
Long-horizon agentic search: Reasoning-Intensive QA

RQ2 Re-ranking helps every retriever, and deeper is better

Answer accuracy (%) after re-ranking passages, gpt-oss-20b

Re-ranker (depth)	BM25	SPLADE-v3	Qwen3-Embed-8B
None	57.2	51.6	41.7
monoT5 (10)	63.1	59.9	47.1
monoT5 (50)	**68.9**	64.6	55.9
RankLLaMA (50)	67.8	**66.3**	**56.8**
Rank1 (50)	68.7	64.3	56.4

- All three re-rankers give similar gains; reasoning (Rank1) shows no clear edge
- BM25 + monoT5 (50) reaches **68.9%**, close to GPT-5 + Qwen3-Embed-8B (70.1%), with fewer search calls



Accuracy vs. search calls, BM25 + re-rankers on passages. Dashed line: GPT-5 + Qwen3-Embed-8B

Meng et al. Revisiting Text Ranking in Deep Research. SIGIR 2026.

Long-horizon agentic search: Reasoning-Intensive QA

RQ3 Rewriting agent queries as questions helps neural rankers

Q2Q turns each agent query into a natural-language question; Q+R also uses the agent's latest reasoning

Example

Query	Text
Raw	"61,880" football attendance
Q2Q (Q)	What is the football attendance number 61,880?
Q2Q (Q+R)	What football match had an attendance of 61,880?

Answer accuracy (%), gpt-oss-20b, passages

Ranker	Raw	Q2Q (Q)	Q2Q (Q+R)
BM25	57.2	**57.8**	55.7
SPLADE-v3	51.6	51.0	**55.7***
Qwen3-Embed-8B	41.7	40.4	**45.9***

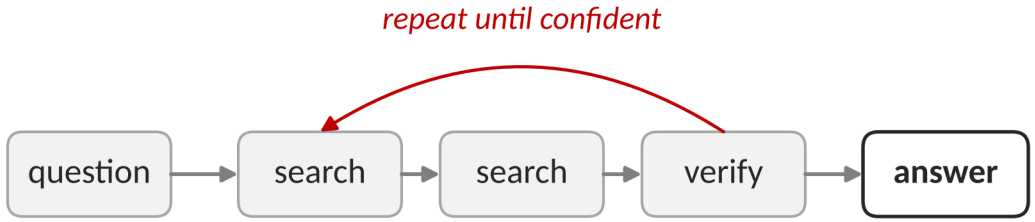
* significant gain over raw queries ($p < 0.05$)

- Using the reasoning keeps the agent's intent; rewriting the query alone can shift it
- Q2Q (Q+R) significantly helps neural retrievers; BM25 does not benefit, as keyword queries already suit it

Long-horizon agentic search: Report generation

Reasoning-intensive QA

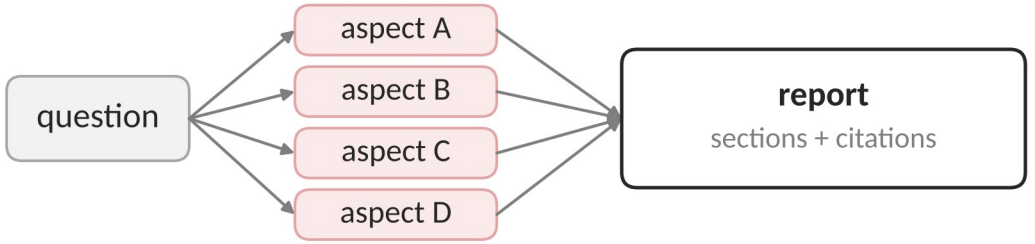
- Short-form, verifiable answers
- Targeted search for specific evidence
- Deep exploration, repeated verification



e.g. *Which fictional character breaks the fourth wall, was helped by selfless ascetics, and had a 1960s–80s TV show with < 50 episodes?* → **Plastic Man**

Report generation

- Broad, open-ended research questions
- No correct answer, no fixed research path
- Decide what to cover, how deep, when to stop



e.g. *Analyse the feasibility of investing in EV charging infrastructure in suburban areas.* → **cited report**

Outline DeepResearch Bench, ResearcherBench → WebWeaver → What benchmarks miss

QA example from BrowseComp: Wei et al. BrowseComp: A Simple Yet Challenging Benchmark for Browsing Agents. arXiv 2025.

Long-horizon agentic search: Report generation

DeepResearch Bench [1] Report quality and citation quality, judged by LLMs

100 PhD-level tasks (50 EN, 50 ZH) · 22 fields · topic mix from 96k real user queries

- Deep research agents write better reports than LLMs with search
- **More citations ≠ more accurate citations**
  - Gemini Deep Research cites by far the most (~165 supported statements per task), at 78% accuracy
  - GPT-4o Search Preview is the most accurate (87%) but supports only ~5 statements per task
- **RACE**: a judge LLM writes weighted criteria per task and scores the report against a reference (50 = on par)
  - Agrees with experts more often than experts agree with each other (71% vs 68% pairwise agreement)
  - The reference matters most: without it, agreement drops to 67%; a plain judge prompt reaches only 59%

System	RACE	Cit. acc. (%)	Eff. cit.
Gemini-2.5-Pro Deep Research	**49.71**	78.30	**165.34**
OpenAI Deep Research	46.45	75.01	39.79
Perplexity Research	40.46	82.63	31.20
Grok Deeper Search	38.22	73.08	8.58
GPT-4o Search Preview	30.74	**86.63**	5.05

[1] Du et al. DeepResearch Bench: A Comprehensive Benchmark for Deep Research Agents. ICLR 2026. Scores from the official leaderboard; agreement figures from the paper.

Long-horizon agentic search: Report generation

ResearcherBench [1] Frontier research questions, judged against expert rubrics

65 AI research questions · 35 subjects · from lab discussions, researcher interviews and forums

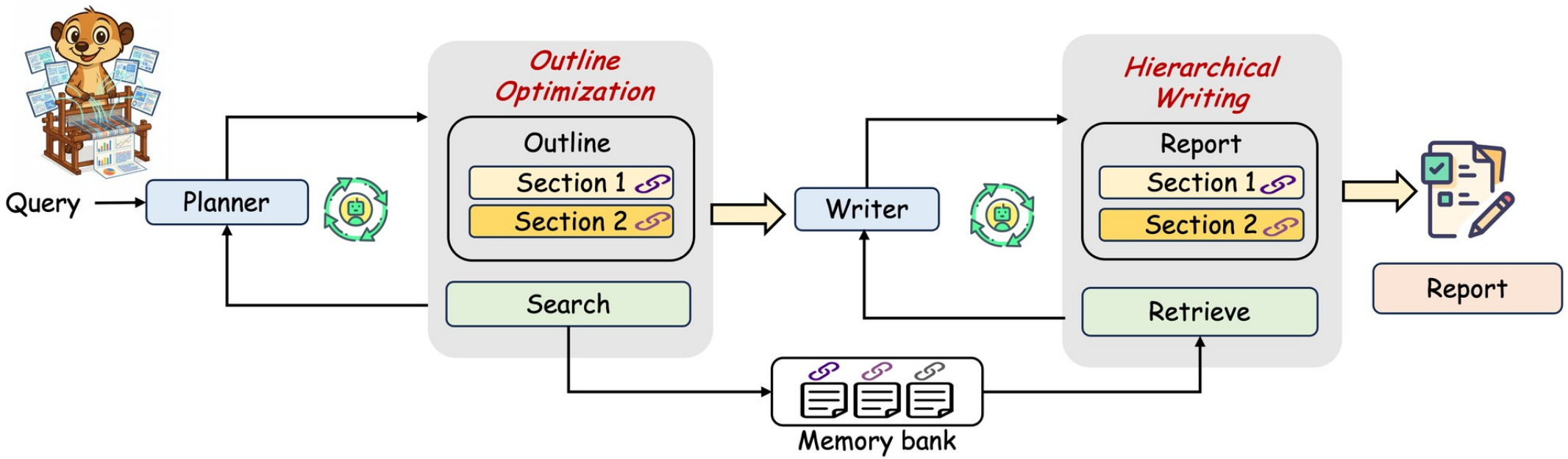
- **Coverage**: experts write rubric items per question (weight 1–3, nice-to-have → core); a judge LLM checks each
  - OpenAI and Gemini Deep Research lead clearly (0.70 and 0.69); no other system passes 0.48
- Cited claims are mostly right, but **most claims are uncited**
  - Faithfulness (supported / cited claims) is 0.84–0.86 for most systems; groundedness (cited / all claims) only 0.32–0.59
  - Systems lean on internal knowledge; citing more ≠ better insight (OpenAI: top coverage, but groundedness only 0.34)
- All systems do best on open consulting (33 of 65 questions), less well on literature review and technical details
  - Top systems exceed 0.75 coverage there: strong ideation partners, weaker as precise literature or technical tools

System	Coverage	Faithful.	Grounded.
OpenAI Deep Research	**0.70**	0.84	0.34
Gemini Deep Research	0.69	**0.86**	**0.59**
Perplexity Deep Research	0.48	0.85	0.56
Grok3 DeepSearch	0.44	0.69	0.32
GPT-4o Search Preview	0.36	**0.86**	0.39

[1] Xu et al. ResearcherBench: Evaluating Deep AI Research Systems on the Frontiers of Scientific Inquiry. arXiv 2025.

Long-horizon agentic search: Report generation

*Both benchmarks reward **coverage, structure and grounded citations**. WebWeaver [1] puts a dynamic outline at the centre of planning and writing.*



- **Planner:** search ⇄ outline; sections cite a memory bank
- **Writer:** section by section, only cited evidence

DeepResearch Bench	RACE	Cit. acc.	Eff. cit.
WebWeaver (Claude-sonnet-4)	**50.58**	**93.37**	**200.75**
Gemini-2.5-Pro Deep Research	49.71	78.30	165.34

[1] Li et al. WebWeaver: Structuring Web-Scale Evidence with Dynamic Outlines for Open-Ended Deep Research. ICLR 2026. Figure and results adapted from [1].

Long-horizon agentic search (deep research)

Q & A

Long-horizon agentic search: Context management

- **Challenges specific to long horizons:** beyond finding the right evidence, long trajectories have to stay tractable
  - **Context management:** preserving useful information across long trajectories (ReSum [1], MEM1 [2], Context-Folding [3])
  - Efficiency: reducing unnecessary searching, reading and reasoning (SlimSearcher, OTC, BATS)

*We have now seen the major tasks, **reasoning-intensive QA and report generation**, and how agents are trained and evaluated for them. Long trajectories also bring difficulties of their own.*

Next: keeping the agent's working context under control

[1] Wu et al. ReSum: Unlocking Long-Horizon Search Intelligence via Context Summarization. 2026.
[2] Zhou et al. MEM1: Learning to Synergize Memory and Reasoning for Efficient Long-Horizon Agents. ICLR 2026.
[3] Sun et al. Scaling Long-Horizon LLM Agent via Context-Folding. ICML 2026.

Long-horizon agentic search: Context management

- **ReAct appends everything:** question → (thought, action, observation) × N
- **Problems:** the history overflows the window (WebSailor-7B passes 32k tokens on hard BrowseComp questions [1]), buries evidence among dead ends, and is re-read at every step
- **Goal:** keep a compact working state with only what the agent needs to reason next

Three strategies that differ in when the context is compressed

ReSum [1]

When the budget is reached

Run ReAct until the context nears its limit, then summarise and restart from the summary

→

MEM1 [2]

Every turn

Rewrite one internal state at each step, merging new evidence and dropping the rest

→

Context-Folding [3]

Proactively, by the agent

Branch into a subtask when the agent chooses; fold it into a short summary on return

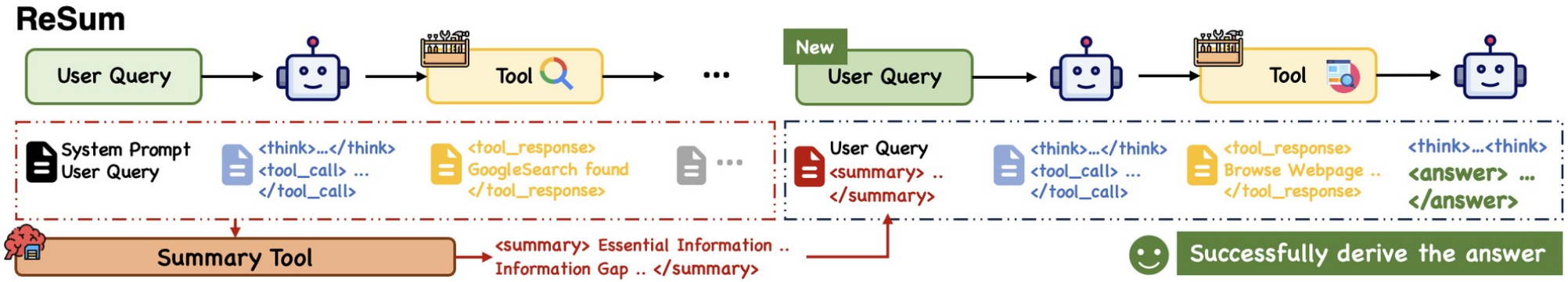
From passive to proactive: the agent state turns from an ever-growing log into a workspace the agent manages

[1] Wu et al. ReSum: Unlocking Long-Horizon Search Intelligence via Context Summarization. 2026.
[2] Zhou et al. MEM1: Learning to Synergize Memory and Reasoning for Efficient Long-Horizon Agents. ICLR 2026.
[3] Sun et al. Scaling Long-Horizon LLM Agent via Context-Folding. ICML 2026.

Long-horizon agentic search: Context management

ReSum [1] Compress when the context budget is reached

- **Trigger:** when the context nears its limit (e.g., 32k tokens), a summary tool condenses the history; the agent restarts from the question plus the summary
- **Summary tool:** ReSumTool-30B, Qwen3-30B-A3B fine-tuned on ~9K conversation–summary pairs, keeps key evidence, flags information gaps and suggests next steps
- **ReSum-GRPO:** split each rollout at its summary points and give every segment the final-answer advantage, so the agent learns to reason from summaries (1K training questions)
- **Results:** training-free ReSum beats ReAct by 4.5% on average (GAIA, BrowseComp-zh, BrowseComp); ReSum-GRPO adds a further 8.2%
- **Trade-off:** plug-and-play for existing ReAct agents, but compression is forced by the budget rather than chosen by the agent

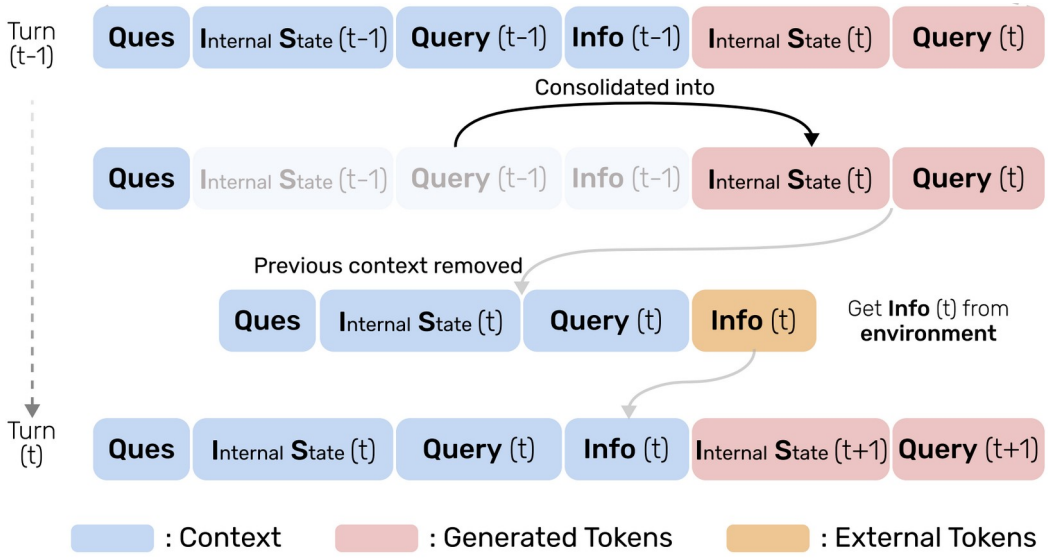


[1] Wu et al. ReSum: Unlocking Long-Horizon Search Intelligence via Context Summarization. 2026. Figure from [1].

Long-horizon agentic search: Context management

MEM1 [1] Compress at every turn

- **Trigger:** every turn, the agent writes a new internal state <IS> that merges its previous state with the latest query and result
- **Prune:** earlier turns are then dropped; the context keeps only the question, the latest <IS>, and the current action and observation
- **Training:** end-to-end RL (PPO) rewarded only on task success; no memory reward, consolidation emerges
- **Masked trajectory:** a 2D attention mask replays each turn's pruned context in one sequence, so policy gradients stay valid
- **Results:** trained on 2-question tasks and tested on 16, MEM1-7B reaches 3.5× the accuracy of Qwen2.5-14B-Instruct with 3.7× lower peak memory
- **Related:** WebResearcher also rebuilds a compact workspace every round [2]

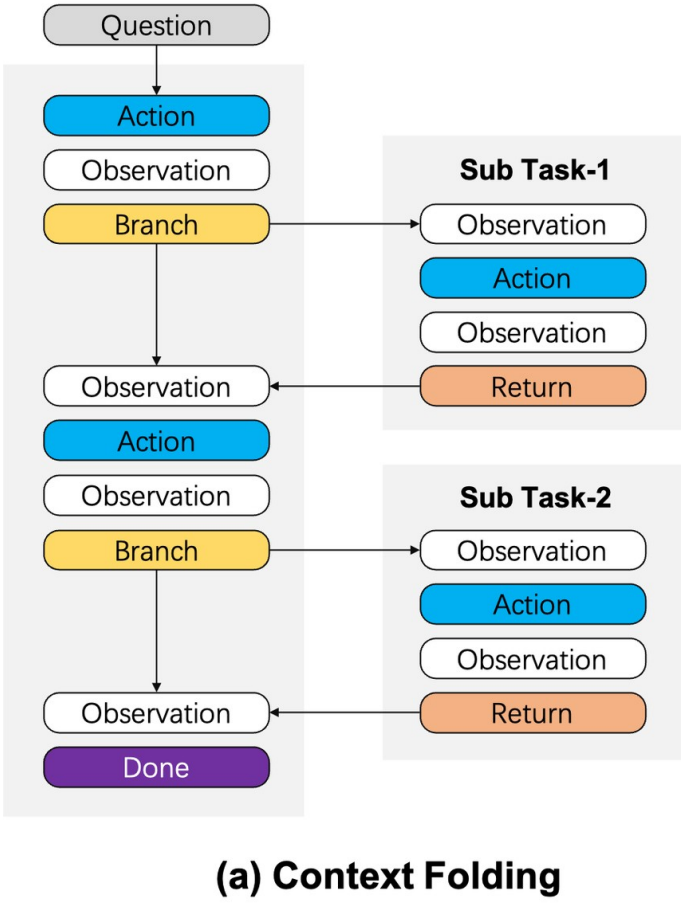


[1] Zhou et al. MEM1: Learning to Synergize Memory and Reasoning for Efficient Long-Horizon Agents. ICLR 2026. Figure from [1].
[2] Qiao et al. WebResearcher: Unleashing Unbounded Reasoning Capability in Long-Horizon Agents. arXiv 2025.

Long-horizon agentic search: Context management

Context-Folding [1] **Compress proactively, when the agent decides**

- **Trigger:** the agent itself decides when to branch into a subtask and what to bring back
- **Two tools:** branch(description, prompt) opens a sub-trajectory for a subtask; return(message) closes it with a short summary of the outcome
- **Fold:** on return, the sub-trajectory's intermediate steps are removed; the main thread keeps only the branch call and the returned summary
- **Working context:** the main thread holds high-level reasoning and outcomes, while long tool outputs and dead ends stay inside branches
- **Result:** matches or beats ReAct on deep research and coding tasks with an active context 10× smaller



(a) Context Folding

[1] Sun et al. Scaling Long-Horizon LLM Agent via Context-Folding. ICML 2026. Figure from [1].

Long-horizon agentic search: Efficiency

- **Challenges specific to long horizons:** beyond finding the right evidence, long trajectories have to stay tractable
  - Context management: preserving useful information across long trajectories (ReSum, MEM1, Context-Folding)
  - **Efficiency:** reducing unnecessary searching, reading and reasoning (SlimSearcher [1], OTC [2], BATS [3])

*Context management keeps long trajectories within the context limit, but it **does not by itself make the agent search less**; by lifting the length ceiling, it can even let agents run more turns.*

Next: doing the same job with fewer search and reasoning steps

[1] Xie et al. SlimSearcher: Training Efficiency-Aware Web Agents via Adaptive Reward Gating. arXiv 2026.
[2] Wang et al. OTC: Optimal Tool Calls via Reinforcement Learning. arXiv 2025.
[3] Liu et al. Budget-Aware Tool-Use Enables Effective Agent Scaling. arXiv 2025.

Long-horizon agentic search: Efficiency

Training time Learn to reach the answer with fewer steps

- **SlimSearcher [1]:** SFT keeps trajectories that are both correct and economical; in RL, reward = correctness gate × tool-call and token multipliers, measured against the cheapest correct rollout in the group
- **OTC [2]:** RL reward scales correctness by how close the tool-call count is to the fewest calls that solved the question
- **Key idea:** efficiency is judged relative to what is achievable for each question, so the agent is not pushed to stop searching too early

SlimSearcher [1] Tongyi-DeepResearch-30B, acc. (%) / rounds

Benchmark	Base	SlimSearcher
GAIA	68.2 / 20.6	**70.9 / 10.6**
BrowseComp	41.0 / 63.7	**44.7 / 47.6**
xbench-DeepSearch	71.3 / 14.3	**79.0 / 5.9**
HLE	35.8 / 23.9	**37.6 / 19.8**

OTC [2] Qwen2.5-7B, EM (%) / tool calls

Benchmark	Search-R1	OTC
NQ	39.9 / 1.70	**44.4 / 0.99**
HotpotQA	34.1 / 2.11	**36.6 / 1.01**

Both GRPO-trained. Across models and settings, OTC cuts tool calls by up to 68.3% at comparable accuracy

[1] Xie et al. SlimSearcher: Training Efficiency-Aware Web Agents via Adaptive Reward Gating. arXiv 2026.
[2] Wang et al. OTC: Optimal Tool Calls via Reinforcement Learning. arXiv 2025.

Long-horizon agentic search: Efficiency

Test time Spend a fixed tool budget wisely, without retraining [1]

- **Budget Tracker:** a prompt-level plug-in that shows used and remaining calls for each tool at every step; plain ReAct stops early and leaves most of a large budget unused
- **BATS:** plans with a checklist of exploration and verification clues; a verifier then decides to dig deeper, pivot or stop, based on the remaining budget
- **Efficiency:** with only 10 calls, Budget Tracker matches ReAct at 100 calls on BrowseComp (12.8% vs 12.6%), with 40% fewer searches and 31% lower cost

Accuracy (%) with a budget of 100 calls per tool

Model / method	BrowseComp	BrowseComp-ZH	HLE-Search
Gemini-2.5-Pro ReAct	12.6	31.5	20.5
Gemini-2.5-Pro + Budget Tracker	14.6	32.9	21.8
Gemini-2.5-Pro BATS	**24.6**	**46.0**	**27.0**
Claude-Sonnet-4 ReAct	12.2	29.1	20.5
Claude-Sonnet-4 + Budget Tracker	14.0	31.1	23.0
Claude-Sonnet-4 BATS	**19.1**	**41.5**	**29.0**

[1] Liu et al. Budget-Aware Tool-Use Enables Effective Agent Scaling. arXiv 2025.

Long-horizon agentic search (deep research)

Q & A

Outline

PART I 14:00–16:00

Basics and optimization

- Introduction to agentic search
- Optimization paradigms
  - No parameter updates
  - Agent learns to search
  - Search learns to serve agents
  - Agent–search joint optimization
- Live demonstration and hands-on

BREAK 16:00–16:30

PART II 16:30–18:30

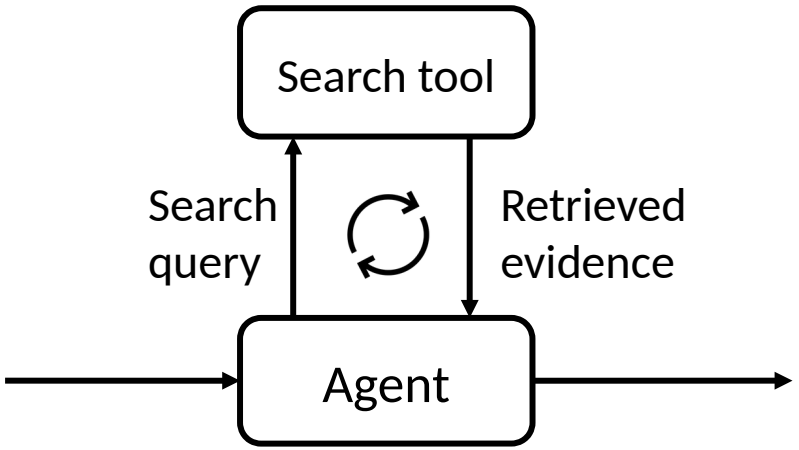
Long-horizons, evaluation and beyond

- Long-horizon agentic search (deep research)
  - Reasoning-intensive QA and report generation
  - Context management
  - Efficiency
- **Evaluation for agentic search**
  - Evaluation for evidence and short answer
  - Evaluation for report
- Multimodal, multilingual and domain-specific
- Conclusion and open challenges

Evaluation for agentic search



I want a day trip from London Saturday this weekend. Find a seaside destination with good weather, less than two hours away by train, and return tickets under £50.



Evidence



Short answer

Brighton

Report

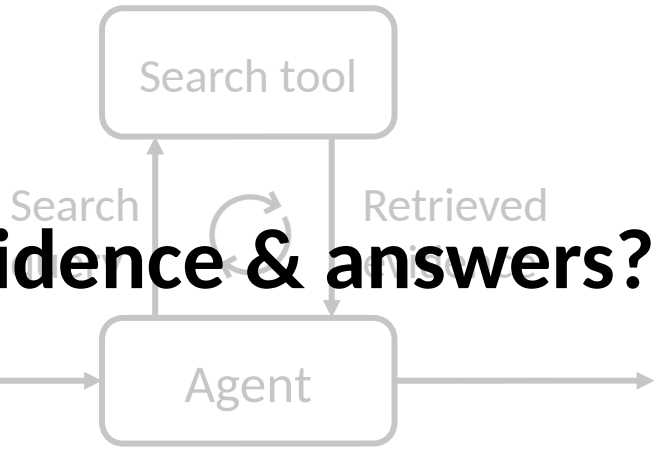
Brighton is sunny, 57 min direct, ~£36 return, with the comparison table:

Candidates	Weather	Train	Return
Brighton	**sunny**	**57 min**	**~£36**
Deal	sunny	1 h 23	~£26
Rye	sunny	1 h 06	~£35
Eastbourne	cloudy	–	–
Durdle Door	cloudy	–	–

Evaluation for agentic search

I want a day trip from London Saturday this weekend. Find a seaside destination with good weather, less than two hours away by train, and return tickets under £50.





How good are the evidence & answers?

Evidence



Short answer

Brighton

Report

Brighton is sunny, 57 min direct, ~£36 return, with the comparison table:

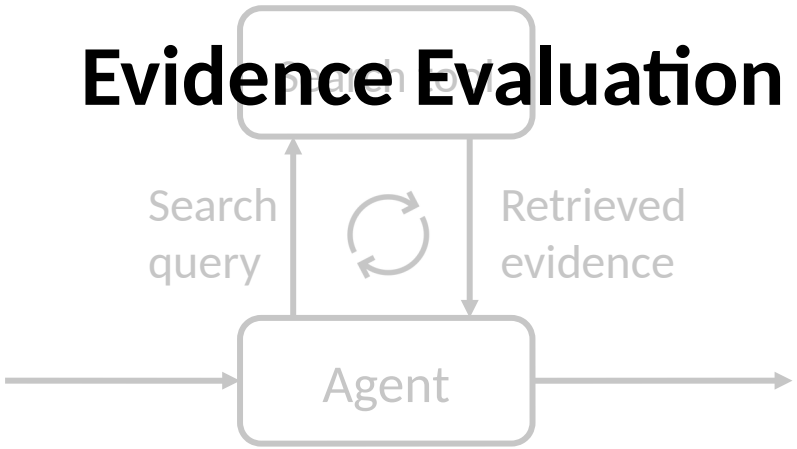
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Durdle Door	cloudy	–	–

Evaluation for agentic search



I want a day trip from London Saturday this weekend. Find a seaside destination with good weather, less than two hours away by train, and return tickets under £50.

Evidence Evaluation



Evidence



Short answer

Brighton

Report

Brighton is sunny, 57 min direct, ~£36 return, with the comparison table:

Candidates	Weather	Train	Return
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Rye	sunny	1 h 06	~£35
Eastbourne	cloudy	–	–
Durdle Door	cloudy	–	–

Evidence Evaluation

- Why? We want responses that are grounded in facts from the corpus, not from the agent's memory.
  - Even if evidence exists in the parametric memory, we want to make sure it's still valid!

- Typical IR systems evaluate ranked lists and (usually) prioritize valuable documents at higher ranks:



  - Based on user browsing behavior: users grow impatient the more they have to read.

Evidence Evaluation

- Agentic search systems are different: multiple sets of search results contribute to a growing context:

```
<input>prompt</input>
<thinking>...</thinking>
<tool_call>search(...)</tool_call>
<tool_response>
```

1.  2.  3.  4.  5. 

```
</tool_response>
<thinking>...</thinking>
<tool_call>search(...)</tool_call>
<tool_response>
```

1.  2.  3.  4.  5. 

```
</tool_response>
...
<output>...</output>
```

Evidence Evaluation

- Agentic search systems are different: multiple sets of search results contribute to a growing context:

```
<input>prompt</input>
<thinking>...</thinking>
<tool_call>search(...)</tool_call>
<tool_response>
```

1. 2. 3. 4. 5.

Documents “read” in parallel

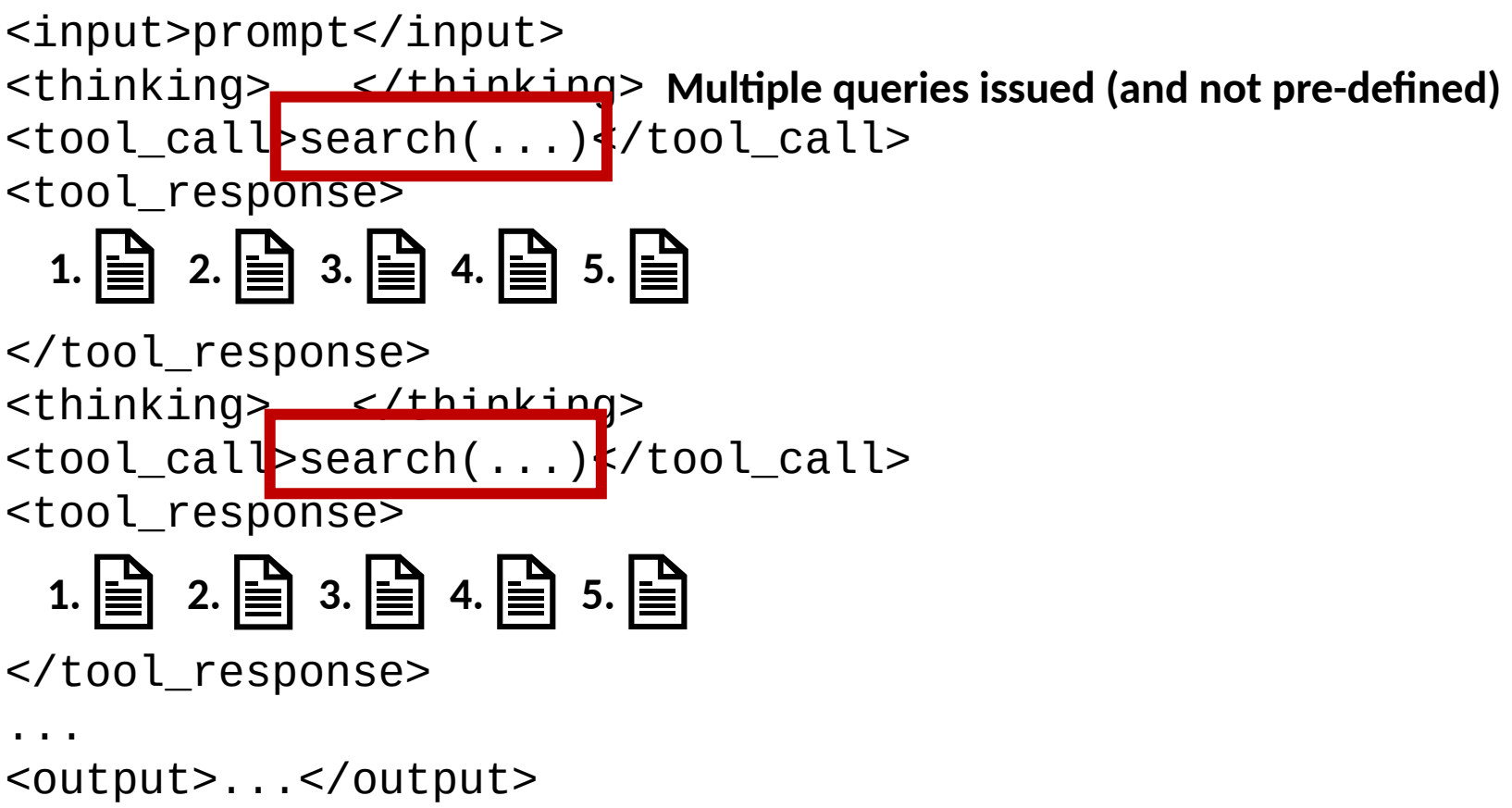
```
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<tool_response>
```

1. 2. 3. 4. 5.

```
</tool_response>
...
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```

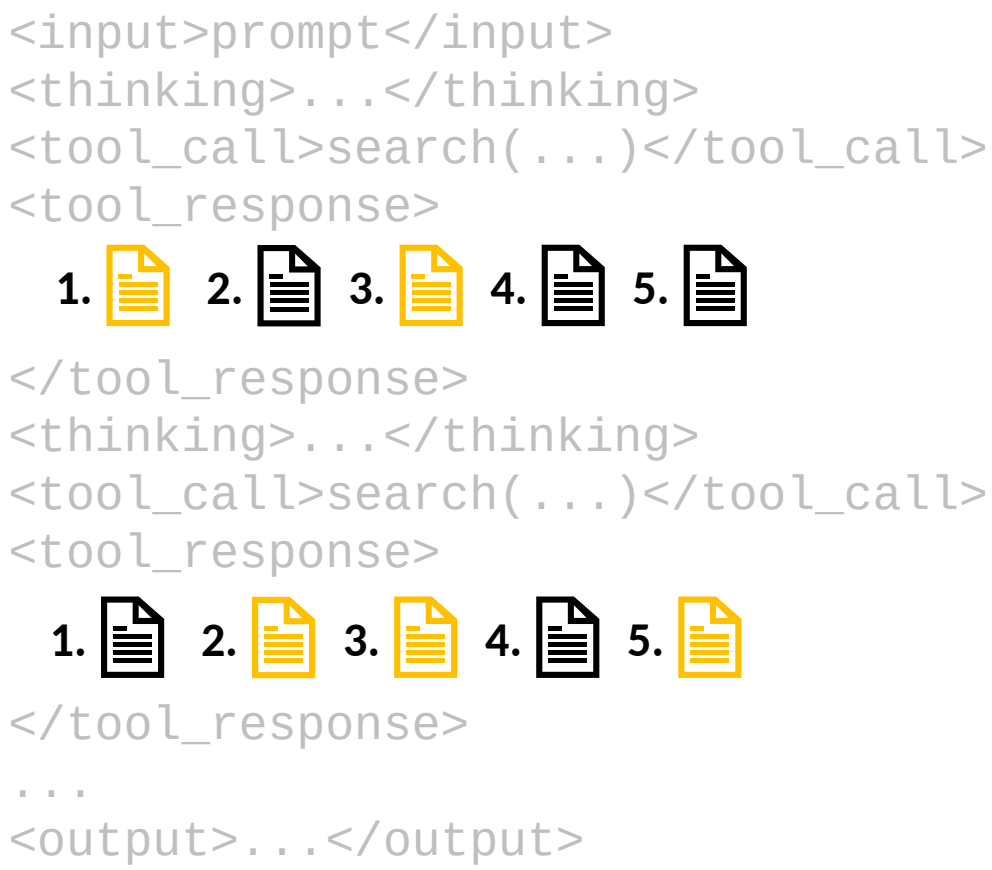
Evidence Evaluation

- Agentic search systems are different: multiple sets of search results contribute to a growing context:



Evidence Evaluation

- Therefore, we often measure evidence as cumulative Recall – the percent of all documents that support an answer retrieved at any stage.



Not retrieved but relevant:

relevant

non-relevant

Recall = 5/8 = 0.625

Evidence Evaluation

- Alternative: subtopic recall – the proportion of subtopics that you retrieved any documents for.
  - Main idea: you don't need *all* evidence for each subtopic, just *some* evidence.

```
<input>prompt</input>
<thinking>...</thinking>
<tool_call>search(...)</tool_call>
<tool_response>
```

1.  2.  3.  4.  5. 

```
</tool_response>
<thinking>...</thinking>
<tool_call>search(...)</tool_call>
<tool_response>
```

1.  2.  3.  4.  5. 

```
</tool_response>
...
<output>...</output>
```

Not retrieved but relevant:

✓ Brighton is near London
✓ Brighton is on the sea
✘ Brighton has good weather
✘ Brighton has cheap trains

ST Recall = 2/4 = 0.5

Evidence Evaluation

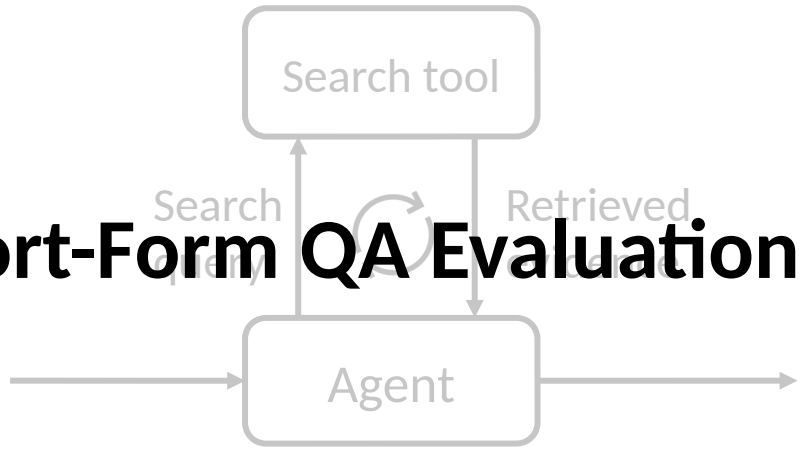
Limitations:

- What counts as a “relevant” document?
  - Human assessed? Contains the final answer? Contains a nugget from the final answer? LLM-Eval?
  - How do we ensure completeness of the judgements (long-standing IR problem)?
- Unbounded recall favors systems that consume lots of results. (More on this later.)
- No proof the agent uses this information, just that it’s exposed to it.
- Evidence recall is not necessarily correlated with downstream effectiveness.

Evaluation for agentic search



I want a day trip from London Saturday this weekend. Find a seaside destination with good weather, less than two hours away by train, and return tickets under £50.



Short-Form QA Evaluation

Evidence



Short answer
Brighton

Report
Brighton is sunny, 57 min direct, ~£36 return, with the comparison table:

Candidates	Weather	Train	Return
Brighton	**sunny**	**57 min**	**~£36**
Deal	sunny	1 h 23	~£26
Rye	sunny	1 h 06	~£35
Eastbourne	cloudy	–	–
Durdle Door	cloudy	–	–

Short-Form Answer Evaluation

- In some settings, a concise answer suffices
- Benefits: direct, simple evaluation
- Limitations: not suitable for all tasks, often not as helpful to the user as long-form responses
- Key challenge: even concise answers can be expressed in many ways:

☑	**Short answer** Brighton
☑	**Short answer** Brighton, UK
☒	**Short answer** Brighton, Queensland
☑	**Short answer** Brighton and Hove
☒	**Short answer** Brighton and Leeds

Short-Form Answer Evaluation

- Classic lexical evaluation:
  - Based on a set of “correct” answers per question…
  - Exact Match (EM) – does the response match an answer exactly?
  - “Accuracy” (sometimes called contains) – does the response contain an answer exactly?
  - F1 – harmonic mean of token-level precision and recall
- Benefits: Fast, deterministic
- Limitations: Still brittle against variations

Known Ground Truth Answers
Brighton
Brighton, UK

Response	EM	Accuracy	F1
☑ Brighton	1.00	1.00	1.00
☑ Brighton, UK	1.00	1.00	1.00
☒ Brighton, Queensland	0.00	1.00	0.67
☑ Brighton and Hove	0.00	1.00	0.50
☒ Brighton and Leeds	0.00	1.00	0.50

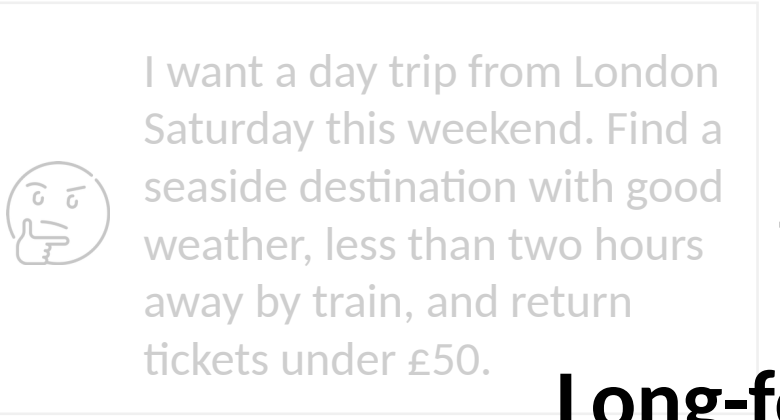
Short-Form Answer Evaluation

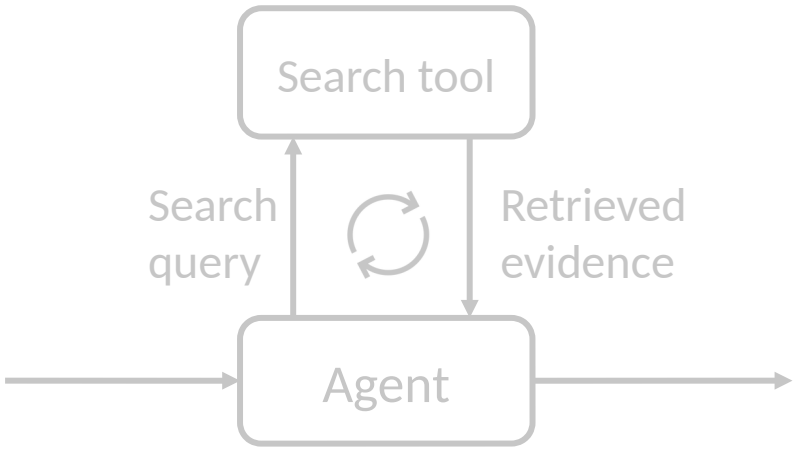
- Model-based evaluation:
  - Also based on a set of “correct” answers per question (but usually just one)…
  - LLM-as-a-judge: Ask an LLM if the response matches the answer(s)
  - BERTScore: Use a trained BERT model to measure answer similarity
- Benefit: More accepting of semantically-equivalent variations
- Limitations: Model noise, slow

Known Ground Truth Answers
Brighton
Brighton, UK

Response	qwen3.8-27b	BERTScore
☑ Brighton	1.00	1.00
☑ Brighton, UK	1.00	1.00
☒ Brighton, Queensland	0.00	0.92
☑ Brighton and Hove	1.00	0.82
☒ Brighton and Leeds	0.00	0.87

Evaluation for agentic search





Evidence



Short answer

Brighton

Report

Brighton is sunny, 57 min direct, ~£36 return, with the comparison table:

Candidates	Weather	Train	Return
Brighton	**sunny**	**57 min**	**~£36**
Deal	sunny	1 h 23	~£26
Rye	sunny	1 h 06	~£35
Eastbourne	cloudy	–	–
Durdle Door	cloudy	–	–

Long-form Report Evaluation

Long-form Report Evaluation

Responses in the format of a “report” offer a more comprehensive answer for the user:

User sees *why* it’s the answer

Report

Brighton is sunny, 57 min direct, ~£36 return, with the comparison table:

Candidates	Weather	Train	Return
Brighton	**sunny**	**57 min**	**~£36**
Deal	sunny	1 h 23	~£26
Rye	sunny	1 h 06	~£35
Eastbourne	cloudy	–	–
Durdle Door	cloudy	–	–

Long-form Report Evaluation

Responses in the format of a “report” offer a more comprehensive answer for the user:

Allows user to draw a different conclusion (when subjective)

Report

Brighton is sunny, 57 min direct, ~£36 return, with the comparison table:

Candidates	Weather	Train	Return
Brighton	**sunny**	**57 min**	**~£36**
Deal	sunny	1 h 23	~£26
Rye	sunny	1 h 06	~£35
Eastbourne	cloudy	–	–
Durdle Door	cloudy	–	–

Long-form Report Evaluation

Responses in the format of a “report” offer a more comprehensive answer for the user:

Report

Brighton is sunny [1], 57 min direct, ~£36 return [2], with the comparison:

Candidates	Weather	Train	Return
Brighton	**sunny**	**57 min**	**~£36**
Deal	sunny	1 h 23	~£26
Rye	sunny	1 h 06	~£35
Eastbourne	cloudy	–	–
Durdle Door	cloudy	–	–

Can link information to sources

Long-form Report Evaluation

Report evaluation is tricky! There are infinitely many equally-valid ways to express the same information.

Report

Brighton is sunny, 57 min direct, ~£36 return, with the comparison table:

Candidates	Weather	Train	Return
Brighton	**sunny**	**57 min**	**~£36**
Deal	sunny	1 h 23	~£26
Rye	sunny	1 h 06	~£35
Eastbourne	cloudy	–	–
Durdle Door	cloudy	–	–

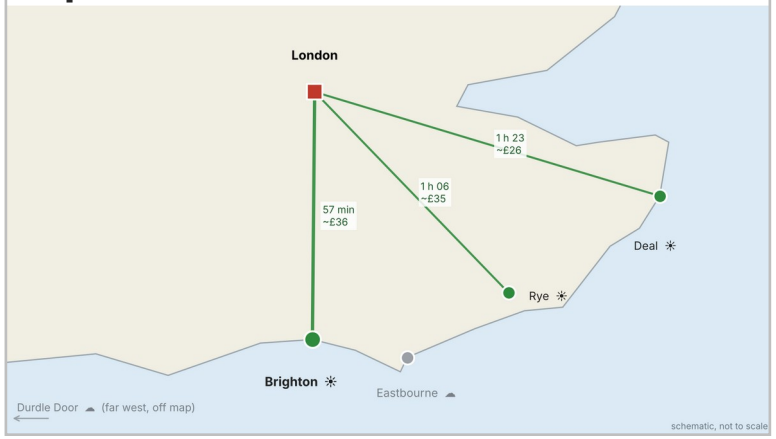
Report

I recommend going to Brighton! It's only a 57 minute train ride away, and will cost you £36 – well within your budget. But you might also want to consider Deal (also nice with a cheaper train journey, but a slightly longer duration) or Rye (about the same duration and cost).

Report

To Brighton, friend, the sun is out,
Fifty-seven minutes, no doubt,
Direct by rail, and for thirty-six,
You'll have your seaside, sand and chips.
Should Deal appeal, the fare is less,
Twenty-six, but I confess
It's one hour twenty-three each way.
Rye's quick too, thirty-five to pay.

Report



Long-form Report Evaluation

Report evaluation often tries to capture:

- **Fact Coverage**: Does the report include the necessary information?
- **Attribution**: Is the information correctly linked to sources that attest it?
- **Text Quality**: Is the information expressed in a coherent way?

Long-form Report Evaluation

Fact Coverage:

- Usually framed as which of a set of gold “nuggets” (atomic facts) are in the report
- These nuggets can be rated based on their importance
- They are sometimes framed as question-answer pairs

#	Nugget	As a question-answer pair	Importance
1	Brighton is a seaside destination reachable from London by train	Is Brighton a seaside destination reachable from London by train? → Yes	vital
2	The train journey to Brighton takes ~57 minutes, direct	What’s the train journey time to Brighton? → ~57 min, direct	vital
3	A return ticket to Brighton costs ~£36	What’s the return fare to Brighton? → ~£36	vital
4	The weather in Brighton on Saturday is sunny	What’s the weather in Brighton on Saturday? → Sunny	vital
5	Deal is sunny, 1 h 23 by train, ~£26 return	Is Deal a viable alternative? → Yes: sunny, 1 h 23, ~£26	okay
6	Rye is sunny, 1 h 06 by train, ~£35 return	Is Rye a viable alternative? → Yes: sunny, 1 h 06, ~£35	okay
7	Eastbourne and Durdle Door are cloudy	Are Eastbourne and Durdle Door viable? → No: cloudy (or too far)	okay

Long-form Report Evaluation

Measuring Fact Coverage:

- Measured in terms of nugget recall (% of gold nuggets supported in a report)
- Human- or LLM-assessed
- Note: longer reports have higher chance of coverage by chance, so length limited (by tokens or chars)

Report
I recommend going to Brighton! It's only a 57 minute train ride away, and will cost you £36 – well within your budget. But you might also want to consider Deal (also nice with a cheaper train journey, but a slightly longer duration) or Rye (about the same duration and cost).

#	Nugget	Importance	Support
1	Brighton is a seaside destination reachable from London by train	vital	Partial
2	The train journey to Brighton takes ~57 minutes, direct	vital	Supported
3	A return ticket to Brighton costs ~£36	vital	Supported
4	The weather in Brighton on Saturday is sunny	vital	Unsupported
5	Deal is sunny, 1 h 23 by train, ~£26 return	okay	Partial
6	Rye is sunny, 1 h 06 by train, ~£35 return	okay	Partial
7	Eastbourne and Durdle Door are cloudy	okay	Unsupported

Vital Coverage = 2/4 = 0.50
All Coverage = 2/7 = 0.29

Long-form Report Evaluation

Measuring Attribution:

- Again, by human assessment or LLM
- Which elements of a report (usually sentences) need a citation?
- If it needs a citation, do the provided citations support the assertions?
- Sentence Support measured by % of cite-worthy sentences that are properly supported.

No citation needed

Report

I recommend going to Brighton! It's only a 57 minute train ride away, and will cost you £36 – well within your budget **[1, 2]**. But you might also want to consider Deal (also nice with a cheaper train journey, but a slightly longer duration) or Rye (about the same duration and cost).

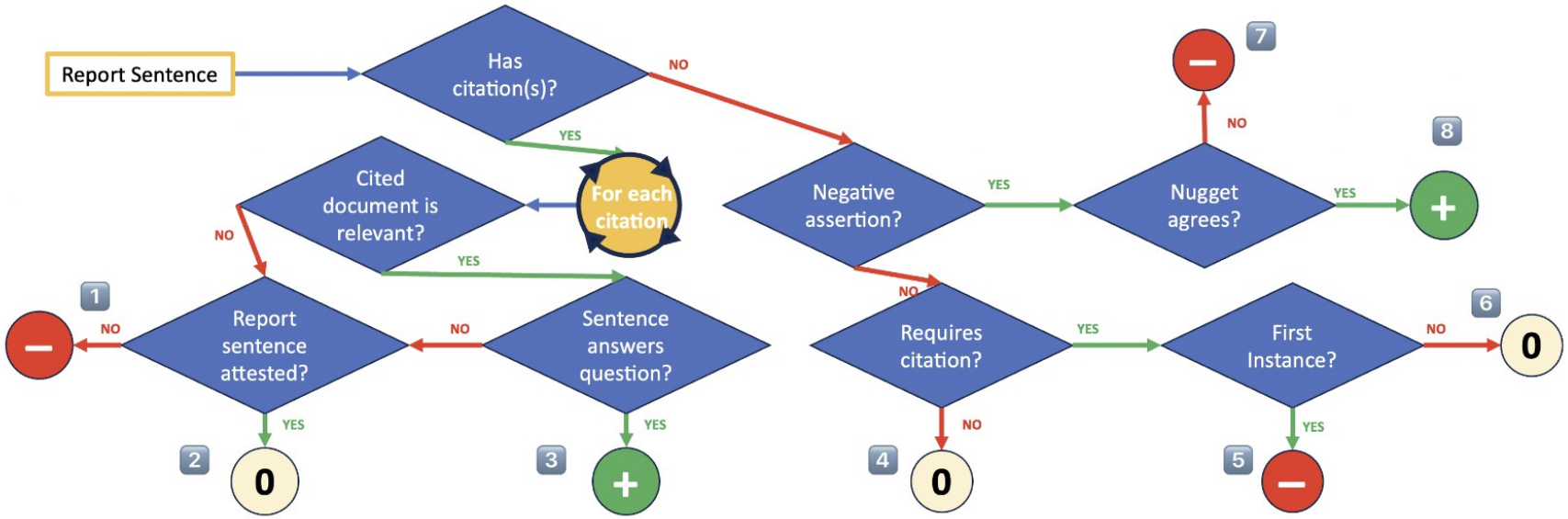
Citation needed and attributed

Citation needed, but not attributed.

Sentence support = 1/2 = 0.50

Long-form Report Evaluation

ARGUE Framework



Mayfield et al. On the Evaluation of Machine-Generated Reports. SIGIR 2024.

Long-form Report Evaluation

Measuring Text Quality:

- Measure things like readability, fluency, narrative structure, etc.
- Usually a set of prompts/rubrics, evaluated by LLM

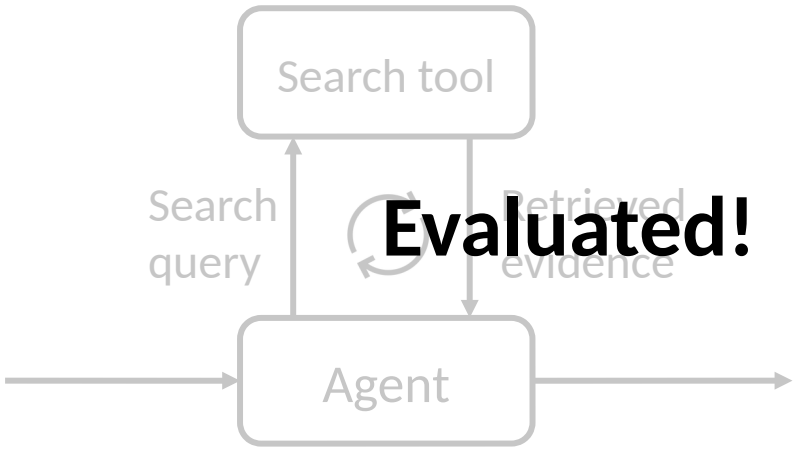
```
…
**Background**: We are evaluating a deep research article written for
the following task across four dimensions: Comprehensiveness, Insight,
Instruction Following, and Readability.
1.  **Comprehensiveness:** The breadth, depth, and relevance of
information coverage.
2.  **Insight:** The depth, originality, logic, and value of the
analysis and conclusions.
3.  **Instruction Following:** Whether the report accurately and
completely responds to all requirements and constraints of the task.
4.  **Readability:** Clarity of structure, fluency of language,
effectiveness of data presentation, and overall ease of understanding.
…
```

Prompt from DeepResearch Bench

Evaluation for agentic search



I want a day trip from London Saturday this weekend. Find a seaside destination with good weather, less than two hours away by train, and return tickets under £50.



Evidence



Short answer

Brighton

Report

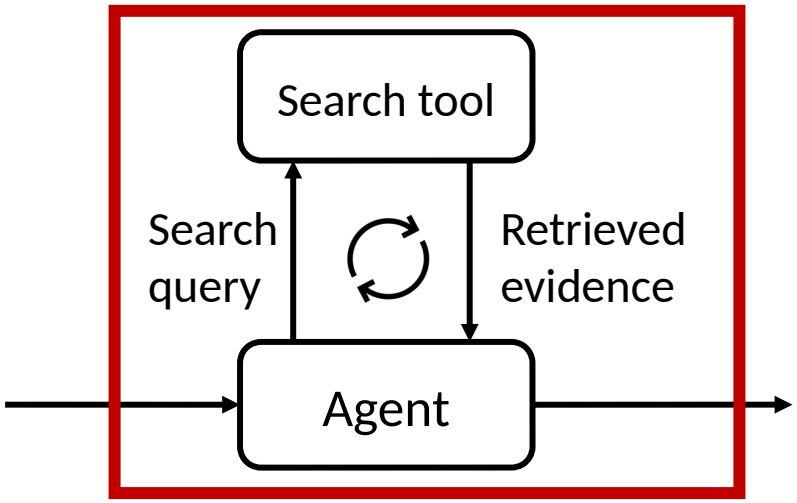
Brighton is sunny, 57 min direct, ~£36 return, with the comparison table:

Candidates	Weather	Train	Return
Brighton	**sunny**	**57 min**	**~£36**
Deal	sunny	1 h 23	~£26
Rye	sunny	1 h 06	~£35
Eastbourne	cloudy	–	–
Durdle Door	cloudy	–	–

Evaluation for agentic search



I want a day trip from London Saturday this weekend. Find a seaside destination with good weather, less than two hours away by train, and return tickets under £50.



Evidence

Short answer

Brighton

Report

Brighton is sunny, 57 min direct, ~£36 return, with the comparison table:

Candidates	Weather	Train	Return
Brighton	**sunny**	**57 min**	**~£36**
Deal	sunny	1 h 23	~£26
Rye	sunny	1 h 06	~£35
Eastbourne	cloudy	–	–
Durdle Door	cloudy	–	–

Bonus: Efficiency Evaluation

Efficiency Evaluation

A system could iterate a long time, to keep searching and collecting more evidence:

Token generation is expensive

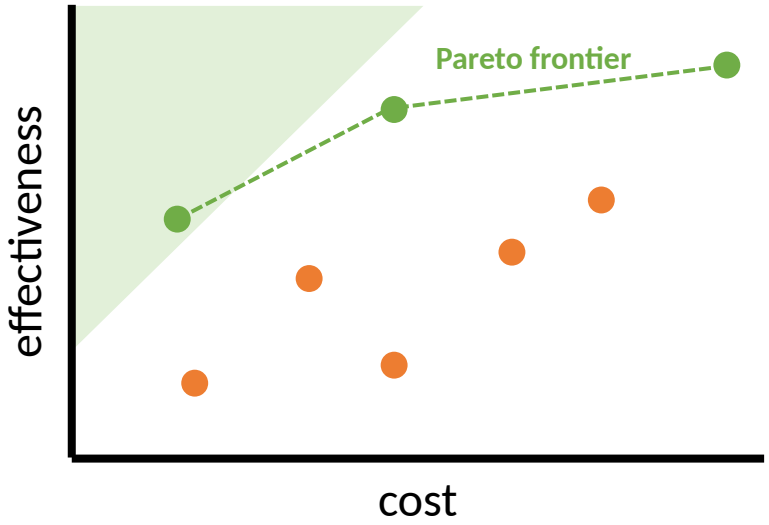
```
<input>prompt</input>
<thinking>...</thinking>
<tool_call>search(...)</tool_call>
<tool_response>...</tool_response>
<thinking>...</thinking>
<tool_call>search(...)</tool_call>
<tool_response>...</tool_response>
<thinking>...</thinking>
<tool_call>search(...)</tool_call>
<tool_response>...</tool_response>
<thinking>...</thinking>
<tool_call>search(...)</tool_call>
<tool_response>...</tool_response>
<thinking>...</thinking>
<tool_call>search(...)</tool_call>
<tool_response>...</tool_response>
<thinking>...</thinking>
<tool_call>search(...)</tool_call>
<tool_response>...</tool_response>
...
<output>...</output>
```

Efficiency Evaluation

A system could iterate a long time, to keep searching and collecting more evidence:

A system that generates fewer tokens is probably preferable to one that generates more
(assuming the same effectiveness)

```
<input>prompt</input>
<thinking>...</thinking>
<tool_call>search(...)</tool_call>
<tool_response>...</tool_response>
<thinking>...</thinking>
<tool_call>search(...)</tool_call>
<tool_response>...</tool_response>
<thinking>...</thinking>
<output>...</output>
```



Efficiency Evaluation

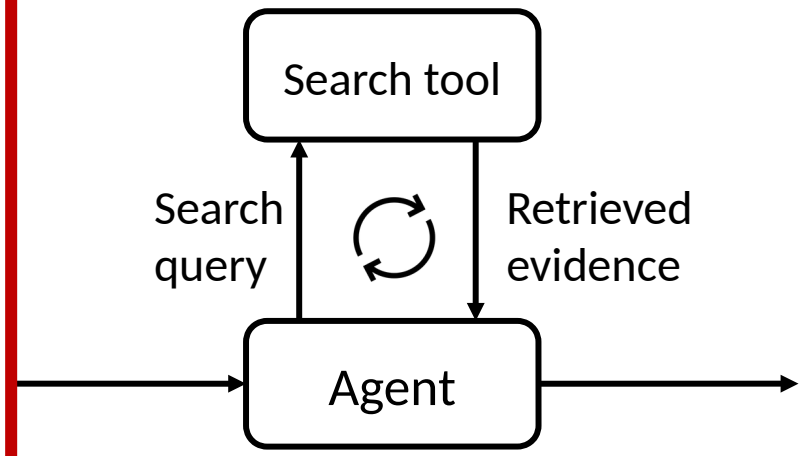
Measuring cost (no measure is perfect)

- **Turns**: the number of search calls invoked (or batches of searches)
  - Common, but really rough estimate of cost
- **Tokens**: the number of input/output tokens of the LLM
  - Some subtlety in measuring this, distinguishing between input/output/cached tokens.
- **Token Cost**: monetary value ($) of the tokens
  - Balances computational cost of input/output/cached tokens
- **Document Reads:** number of documents added to the context
  - Roughly correlated with number of input tokens, since on long chains, this dominates
- **Latency**: wall-clock duration to process the entire request
  - Dependent on hardware+model

Evaluation for agentic search

I want a day trip from London Saturday this weekend. Find a seaside destination with good weather, less than two hours away by train, and return tickets under £50.

Evaluated!



Evidence



Short answer

Brighton

Report

Brighton is sunny, 57 min direct, ~£36 return, with the comparison table:

Candidates	Weather	Train	Return
Brighton	**sunny**	**57 min**	**~£36**
Deal	sunny	1 h 23	~£26
Rye	sunny	1 h 06	~£35
Eastbourne	cloudy	–	–
Durdle Door	cloudy	–	–

Evaluation for agentic search

Tools

- https://github.com/hltcoe/auto-argue
- https://github.com/castorini/nuggetizer
- https://github.com/manestay/dogmatiq
- https://github.com/trec-auto-judge/auto-judge-evaluate

Bibliography

Voorhees. Overview of the TREC 2003 Question Answering Track. TREC 2003

Zhai et al. Beyond independent relevance: methods and evaluation metrics for subtopic retrieval. SIGIR 2003

Mayfield et al. On the Evaluation of Machine-Generated Reports. SIGIR 2024

Walden et al. Auto-ARGUE: LLM-Based Report Generation Evaluation. SIGIR 2026

Li et al. DoGMaTiQ: Automated Generation of Question-and-Answer Nuggets for Report Evaluation. ICTIR 2026

Pradeep et al. The Great Nugget Recall: Automating Fact Extraction and RAG Evaluation with Large Language Models. SIGIR 2025.

Du et al. DeepResearch Bench: A Comprehensive Benchmark for Deep Research Agents. ICLR 2026

Chen et al. BrowseComp-Plus: A More Fair and Transparent Evaluation Benchmark of Deep-Research Agent. arXiv 2025.

Farzi and Dietz. EXAM++: LLM-based Answerability Metrics for IR Evaluation. LLM4Eval 2024.

Namuduri et al. GENIE: A Fine-Grained Measure for Novelty. EMNLP 2026.

Evaluation for agentic search

Q & A

Outline

PART I 14:00–16:00

Basics and optimization

- Introduction to agentic search
- Optimization paradigms
  - No parameter updates
  - Agent learns to search
  - Search learns to serve agents
  - Agent–search joint optimization
- Live demonstration and hands-on

BREAK 16:00–16:30

PART II 16:30–18:30

Long-horizons, evaluation and beyond

- Long-horizon agentic search (deep research)
  - Reasoning-intensive QA and report generation
  - Context management
  - Efficiency
- Evaluation for agentic search
  - Evaluation for evidence and short answer
  - Evaluation for report
- **Multimodal, multilingual and domain-specific**
- Conclusion and open challenges

Multimodal, multilingual and domain-specific

- **Beyond English text on the open web:** most agentic search work assumes evidence is textual, in the language of the query, and on the open web; real-world tasks often break these assumptions
  - **Multimodal (images and video):** WebWatcher [1], Deep Video Discovery [2], VideoDR [3]
  - **Cross-lingual:** XBCP [4]
  - **Domain-specific (legal, medical, enterprise):** L-MARS [5], DeepMed [6], OfficeQA Pro [7]

*The loop stays the same; what changes is **what the agent searches over** and **which tools it needs**.*

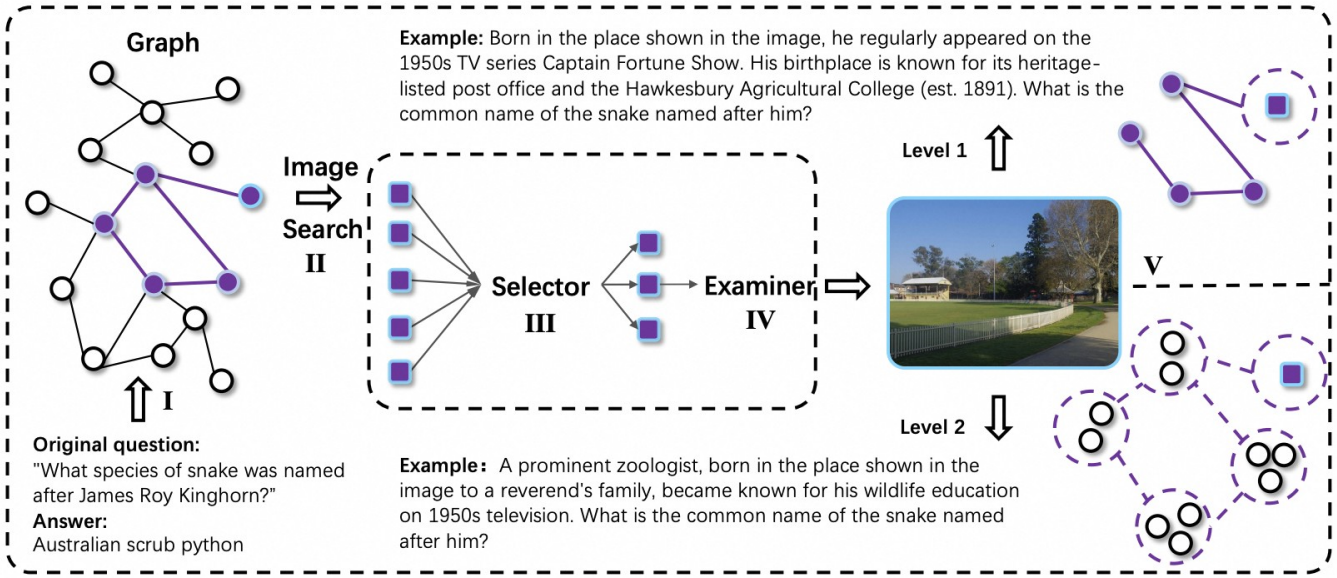
Next: multimodal agentic search over images with WebWatcher

[1] Geng et al. WebWatcher: Breaking New Frontiers of Vision-Language Deep Research Agent. ICLR 2026.
[2] Zhang et al. Deep Video Discovery: Agentic Search with Tool Use for Long-form Video Understanding. NeurIPS 2025.
[3] Liu et al. Watching, Reasoning, and Searching: A Video Deep Research Benchmark on Open Web for Agentic Video Reasoning. arXiv 2026.
[4] Lu et al. Beyond Monolingual Deep Research: Evaluating Agents and Retrievers with Cross-Lingual BrowseComp-Plus. arXiv 2026.
[5] Wang & Yuan. L-MARS: Legal Multi-Agent Workflow with Orchestrated Reasoning and Agentic Search. arXiv 2025.
[6] Wang et al. DeepMed: Building a Medical DeepResearch Agent via Multi-hop Med-Search Data and Turn-Controlled Agentic Training & Inference. Findings of ACL 2026.
[7] Databricks AI Research. OfficeQA Pro: An Enterprise Benchmark for End-to-End Grounded Reasoning. arXiv 2026.

Multimodal agentic search: Images

WebWatcher learns when to search with text and when with images

- **Text-only search agents** cannot use the image; **vision-only agents** over-analyse it and compound errors
- **Data construction:**
  - multi-hop QA from random walks over web knowledge; Level 2 obfuscates entities
  - an image found by image search replaces the key entity
  - selector drops leaky or irrelevant images; examiner keeps queries answerable from the images
- **Training:**
  - cold-start SFT on synthetic multimodal tool-use trajectories
  - RL (GRPO) to improve generalisation

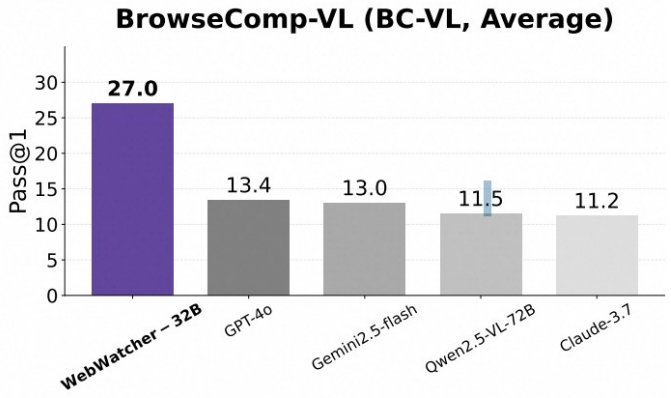


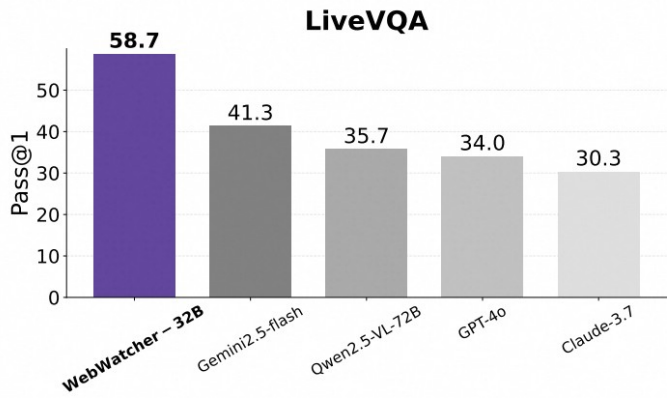
Data construction: graph random walk (I) → image search (II) → selector (III) → examiner (IV) → Level 1 / Level 2 questions (V)

Geng et al. WebWatcher: Breaking New Frontiers of Vision-Language Deep Research Agent. ICLR 2026.

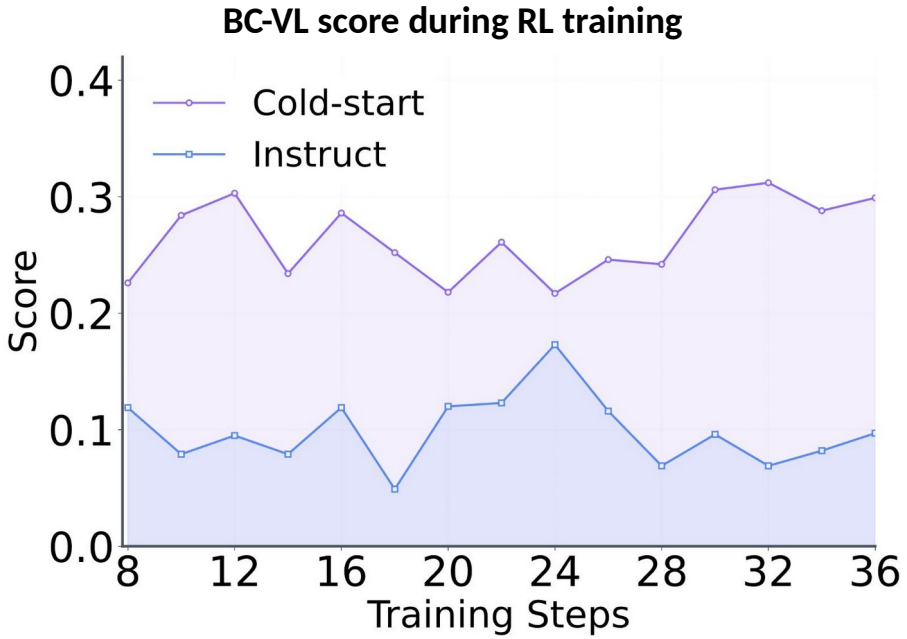
Multimodal agentic search: Images

WebWatcher agentic search beats RAG, and tool use adapts to the task





Pass@1 (%)	BC-VL	LiveVQA	MMSearch
GPT-4o direct	5.5	29.7	18.7
GPT-4o + RAG	13.4	34.0	24.1
WebWatcher-32B	**27.0**	**58.7**	**55.3**



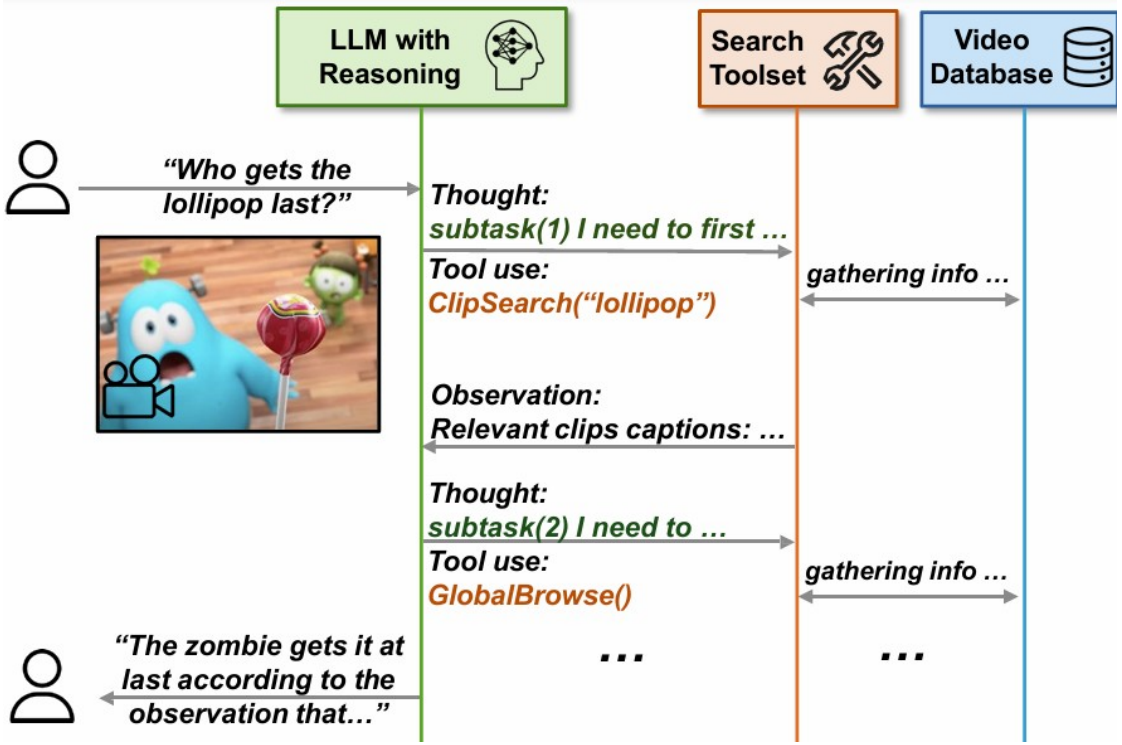
- **Search, then agency:** RAG lifts GPT-4o on BrowseComp-VL from 5.5 to 13.4; WebWatcher's agent loop reaches 27.0
- **Cold start is essential:** after SFT on tool-use trajectories, RL reaches ~0.31; from the instruct model it stays below ~0.17, as tool-format errors wipe out the reward

Geng et al. WebWatcher: Breaking New Frontiers of Vision-Language Deep Research Agent. ICLR 2026.

Multimodal agentic search: Long-form video

Deep Video Discovery [1] search the video instead of watching all of it

- **Index:** split the video into clips; store captions and frames
- **Search:** tools at three levels of detail: skim summaries of the whole video, retrieve the clips whose captions best match a query, and ask a vision model about the raw frames in a chosen time window
- **Stop:** answer once the gathered evidence is enough
- **74.2%** on LVBench [2] (hour-long videos), vs. 60.8% for the previous best agent



The agent alternates thoughts and tool calls until it can answer

[1] Zhang et al. Deep Video Discovery: Agentic Search with Tool Use for Long-form Video Understanding. NeurIPS 2025.
[2] Wang et al. LVBench: An Extreme Long Video Understanding Benchmark. ICCV 2025.

Multimodal agentic search: Video deep research

VideoDR [1] visual clues from the video, answers from the open web

- Videos give only **local visual clues**; the verifiable answer lives on the web
- Extract cross-frame **visual anchors** → turn them into queries → **multi-hop web search** → verify
- e.g. identify a museum from the video, then find the "don't miss" exhibit closest to the entrance and return its accession number
- **500** human-annotated questions across six domains
- **Agentic is not consistently better than Workflow:** early visual anchors drift over long search chains

Accuracy (%) on VideoDR

Model	Workflow	Agentic
Gemini-3.1-pro-preview	51.8	53.6
GPT-5.4	47.0	47.4
Qwen3.5-35B-A3B	35.0	34.6
Gemma-4-31B-it	31.6	33.6
GLM-4.6V-Flash	17.8	21.8
InternVL3.5-14B	16.0	15.4
Human (reference)	40.2	40.2

Workflow: an MLLM first turns video clues into text, then searches without the video. Agentic: one MLLM sees the raw video and searches in a loop.

[1] Liu et al. Watching, Reasoning, and Searching: A Video Deep Research Benchmark on Open Web for Agentic Video Reasoning. arXiv 2026.

Cross-lingual agentic search: XBCP

XBCP [1] English questions, evidence in 12 languages

- BrowseComp-Plus questions and answers stay in English; the **evidence documents are translated** into 11 other languages (high-resource and low-resource)
- **Cross-lingual:** each query's evidence is in one assigned language; Multilingual**:** evidence spread across all languages

Retrieval side: recall and accuracy drop

Corpus	Accuracy	Evidence recall
Original (EN)	64.7	72.8
Multilingual	48.8	59.8
Cross-lingual	42.3	53.8

DeepSeek-V4-Pro agent + Qwen3-Embedding-8B retriever

Agent side: accuracy drops even with all gold evidence

Agent	Original	Multiling.	Cross-ling.
GPT-OSS-20B	90.4	81.2	77.6
GPT-OSS-120B	94.7	89.5	85.3
Qwen3.6-35B-A3B	93.9	90.0	89.2

Accuracy (%) when the agent is given all gold evidence

Two bottlenecks: retrievers miss translated evidence, and agents struggle to integrate it; they also become less calibrated and cite evidence less reliably.

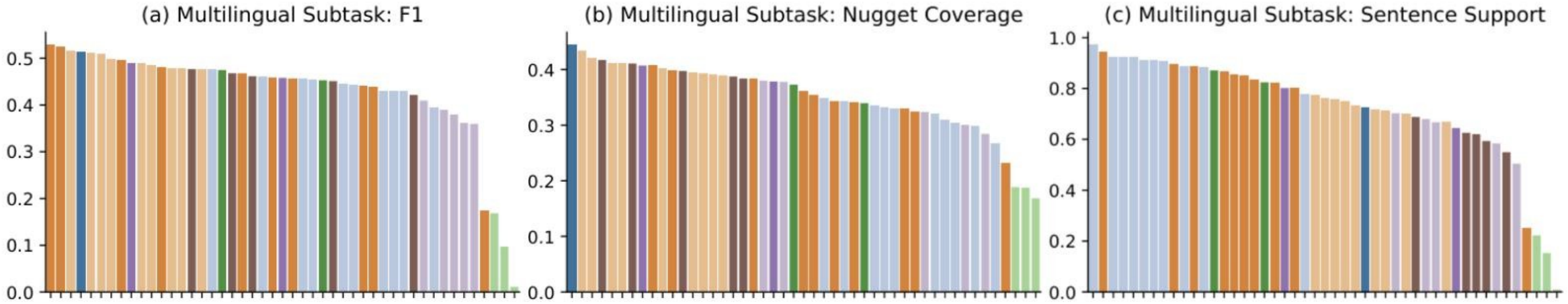
[1] Lu et al. Beyond Monolingual Deep Research: Evaluating Agents and Retrievers with Cross-Lingual BrowseComp-Plus. arXiv 2026.

Multilingual agentic search: TREC RAGTIME

RAGTIME [1] cited English reports from Arabic, Chinese, English and Russian news

- **Task:** given a report request (problem statement + user background), search ~4M news articles (1M each in Arabic, Chinese, English and Russian) and write a **cited English report** of 2,000 or 10,000 characters
- **Scoring (ARGUE): nugget coverage** = share of the assessors' questions the report answers; **sentence support** = share of cited sentences backed by the cited document. 13 teams, 125 runs in 2025

Multilingual report generation, 47 runs: well grounded, but no run covers half the nuggets



Grounding is nearly solved with simple checks; coverage is not. Diverse multilingual retrieval is the main lever.

[1] Lawrie et al. Overview of the TREC 2025 RAGTIME Track. TREC 2025 (arXiv:2602.10024). trec-ragtime.github.io

Domain-specific agentic search: Legal

L-MARS [1] legal answers need current, jurisdiction-specific evidence

Query agent
Breaks down the question; extracts jurisdiction and date

→

Search agent
Web, local legal documents, case law

→

Judge agent
Enough, in-jurisdiction and current? If not: refine and search again

→

Summary agent
Writes an answer with citations

LegalSearchQA accuracy (%)

Method	Accuracy
Zero-shot LLM	58.0
Chain-of-thought	30.0
L-MARS	**96.0**

- Recent legal changes are not in the model's parameters, so **search is necessary**
- Reasoning longer without evidence **makes things worse** (58.0 → 30.0)
- On Bar Exam QA, where the law is stable, search adds only **+0.7**

[1] Wang & Yuan. L-MARS: Legal Multi-Agent Workflow with Orchestrated Reasoning and Agentic Search. arXiv:2509.00761.

Domain-specific agentic search: Medical

DeepMed [1] general deep research agents find medical evidence but fail to use it

- **Gap:** clinical reasoning, not retrieval, limits general DR agents; more tool calls add noisy context
- **Data:** multi-hop med-search QA built from authoritative medical websites
- **Training:** agentic SFT, then RL with a difficulty-aware turn penalty
- **Inference:** an over-evidence monitor stops searching once the answer stops changing

Accuracy (%), selected benchmarks

Model	HLE-Med	MedXpert	MedMCQA	CMExam
Qwen3-14B (base)	12.8	22.4	68.2	82.8
Qwen3-30B-A3B-Thinking	14.8	26.4	73.1	83.8
Tongyi-DeepResearch-30B	28.2	35.4	75.2	87.6
DeepMed-14B (RL)	**26.8**	**36.1**	**82.6**	**91.2**

Across seven medical benchmarks, DeepMed improves its base model by 9.79% on average.

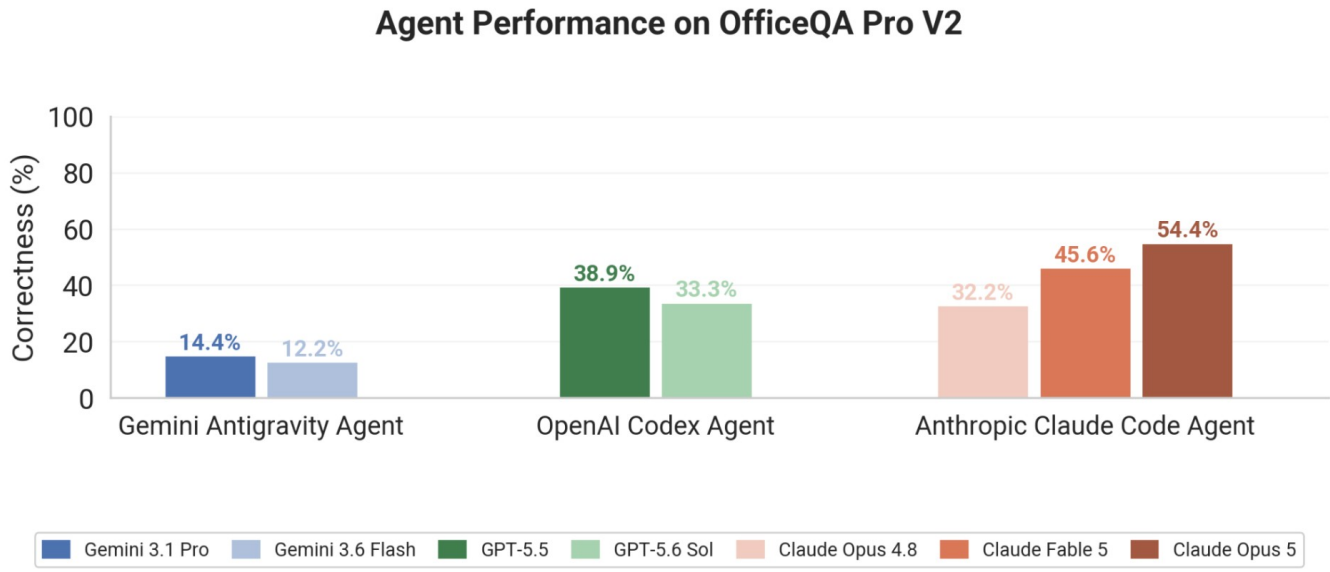
Domain sources are not enough on their own: the agent also needs domain reasoning to use what it finds.

[1] Wang et al. DeepMed: Building a Medical DeepResearch Agent via Multi-hop Med-Search Data and Turn-Controlled Agentic Training & Inference. Findings of ACL 2026.

Domain-specific agentic search: Enterprise

OfficeQA Pro [1] grounded reasoning over enterprise document collections

- Evidence sits in **large private corpora of dense PDFs** (wide tables, scans, legacy layouts)
- Agents must **parse, retrieve and compute** across documents
- **Pro:** 133 questions over U.S. Treasury Bulletins (1939–2025)
- **Pro V2 [2]:** 90 questions over 1,435 federal receipts and expenditures documents (1793–2024); median 5.5 source documents per question
- Best agent harness: **54.4%**



End-to-end correctness on Pro V2: agents use file search (read, grep, glob), web search and code execution.

[1] Databricks AI Research. OfficeQA Pro: An Enterprise Benchmark for End-to-End Grounded Reasoning. arXiv 2026.
[2] Databricks. Introducing OfficeQA Pro V2: A New Benchmark for Enterprise Grounded-Reasoning. 2026.

Multimodal, multilingual and domain-specific

Q & A

Outline

PART I 14:00–16:00

Basics and optimization

- Introduction to agentic search
- Optimization paradigms
  - No parameter updates
  - Agent learns to search
  - Search learns to serve agents
  - Agent–search joint optimization
- Live demonstration and hands-on

BREAK 16:00–16:30

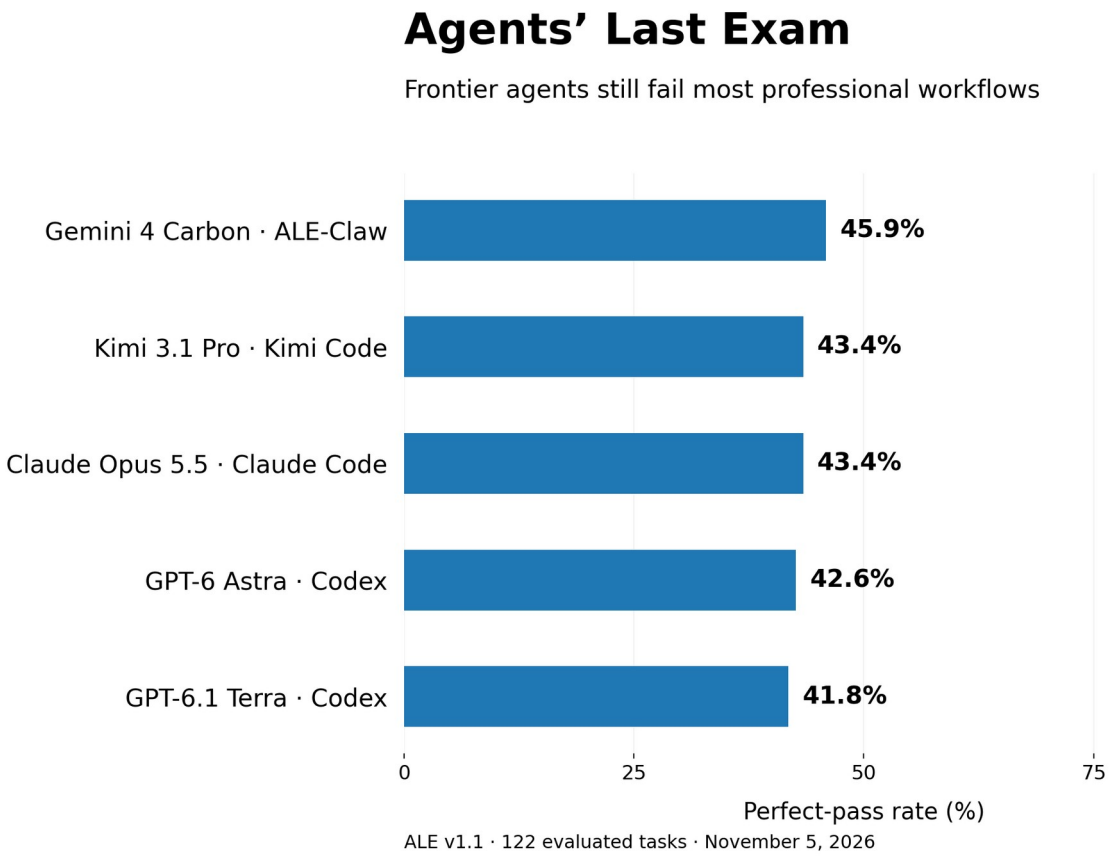
PART II 16:30–18:30

Long-horizons, evaluation and beyond

- Long-horizon agentic search (deep research)
  - Reasoning-intensive QA and report generation
  - Context management
  - Efficiency
- Evaluation for agentic search
  - Evaluation for evidence and short answer
  - Evaluation for report
- Multimodal, multilingual and domain-specific
- **Conclusion and open challenges**

Open Challenges

- **Realistic Tasks and Datasets**
  - Open-ended, real-world scenarios that run for days
  - Realistic environments for long-horizon research
- **Evaluation of Agents**
  - Beyond text: multimodal evidence and generation
  - Long-horizon evaluation: progress, error recovery, and adaptation
- **Retrieval for Agents**
  - Agent-specific query formulation and languages
  - Rankers optimized for agentic search
- **Design and Development of Research Agents**
  - Harness design and agentic search architectures
  - Adaptive workflows and proactive search agents



Conclusion

Key takeaways

- **Agentic search goes beyond traditional retrieval:** agents iteratively reason, search, and refine their answers
- **Optimization spans both agents and search systems:** from agent learning to retrieval and joint optimization
- **Long-horizon research brings new challenges:** evidence management, efficiency, and evaluation across diverse tasks and modalities

Looking ahead

- From benchmark QA to **realistic, open-ended research tasks**
- From traditional retrieval to **IR systems designed for agents**
- From fixed workflows to **adaptive and proactive research agents**